

Highly aneuploid non-small cell lung cancer shows enhanced responsiveness to concurrent radiation and immune checkpoint blockade

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Over 500 clinical trials are investigating combination radiotherapy and immune checkpoint blockade (ICB) as cancer treatments; however, the majority of trials have found no positive interaction. Here we perform a comprehensive molecular analysis of a randomized phase I clinical trial of patients with non-small cell lung cancer (NSCLC) treated with concurrent or sequential ablative radiotherapy and ICB. We show that concurrent treatment is superior to sequential treatment in augmenting local and distant tumor responses and in improving overall survival in a subset of patients with immunologically cold, highly aneuploid tumors, but not in those with less aneuploid tumors. In addition, radiotherapy alone decreases intratumoral cytotoxic T cell and adaptive immune signatures, whereas radiotherapy and ICB upregulates key immune pathways. Our findings challenge the prevailing paradigm that local ablative radiotherapy beneficially stimulates the immune response. We propose the use of tumor aneuploidy as a biomarker and therapeutic target in personalizing treatment approaches for patients with NSCLC treated with radiotherapy and ICB.

Lung cancer remains the leading cause of cancer-related deaths in the United States. Immune checkpoint blockade (ICB) has revolutionized the management of metastatic non-small cell lung cancer (NSCLC) lacking targetable driver mutations; however, only a fraction of patients respond to treatment¹. Mounting preclinical evidence indicates that radiotherapy can exert immunomodulatory effects to augment immunotherapeutic responses^{2–6}. There is also evidence to suggest that anti-programmed cell death protein 1 (anti-PD-1) and anti-cytotoxic

T-lymphocyte-associated protein 4 (anti-CTLA-4) immune checkpoint inhibitors enhance both local and distant tumor responses to radiotherapy⁷. The potential synergy between radiotherapy and ICB has led to over 500 clinical trials in various malignancies, based on the hypothesis that there is a synergistic antitumor interaction between radiotherapy and ICB. Although there is evidence that consolidation ICB after radiotherapy can improve outcomes in some settings^{8,9}, this has also been demonstrated after surgery and therefore does not prove

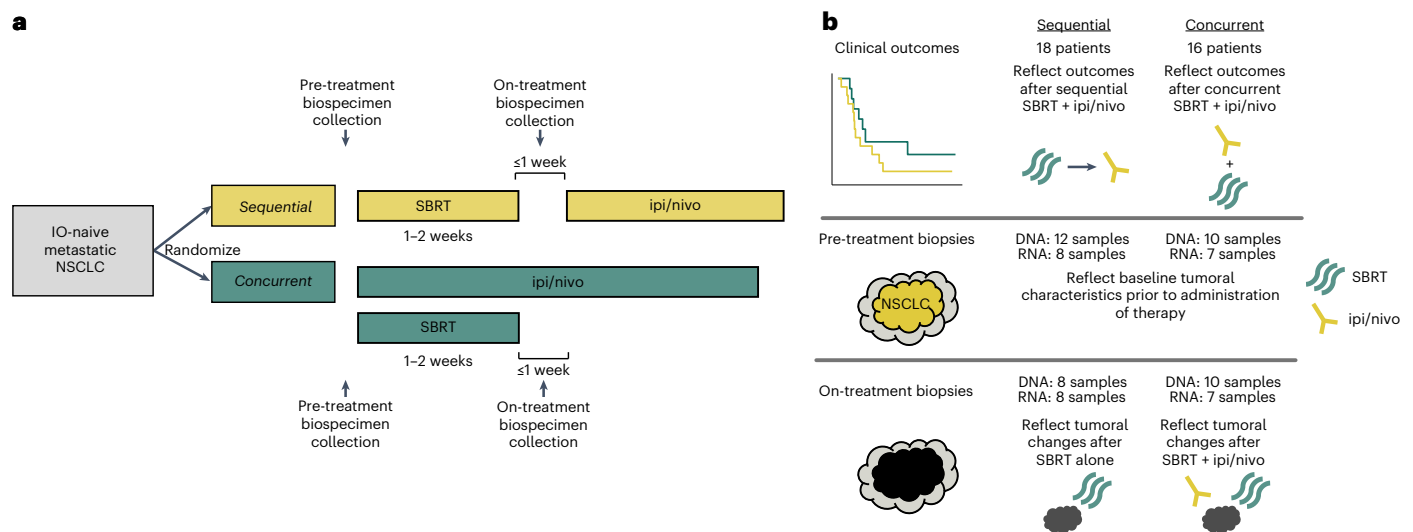


Fig. 1 | Schematic of clinical trial. a, Randomized phase I COSINR clinical trial design. **b**, Schematic demonstrating the differences in biopsy time points by treatment arm and their implications for interpretation of clinical, genomic and transcriptomic results. IO, immunotherapy.

the existence of a synergistic interaction of ICB and radiotherapy. Furthermore, limited biological evidence exists in patients to optimize the potential interaction between local radiotherapy and the immune system with regard to radiation dose, fractionation, timing, and target organs.

At present, a critical clinical question is whether and how radiotherapy and ICB can be combined in a manner that will improve outcomes for patients. Moreover, identification of biomarkers of response will be instrumental in selecting patients who are most likely to benefit from combination therapy. Currently, programmed death-ligand 1 (PD-L1) expression, tumor mutational burden (TMB), effector T cell gene signature expression and the number of tumor neoantigens have been demonstrated to predict anti-PD-1 and/or anti-CTLA-4 response^{10–14}. However, it remains unknown whether any of these or other biomarkers predict responses to the combination of radiotherapy and ICB.

We conducted a randomized clinical trial designed to evaluate the safety and efficacy of combination ICB plus multisite ablative radiotherapy as a first-line treatment for patients with biomarker-unselected metastatic NSCLC. As a secondary analysis of the trial, we present a comprehensive molecular characterization of matched pretreatment and on-treatment tumor biopsies in this unique dataset to delineate the effect of radiotherapy and ICB timing on the tumor immunogenomic milieu. We define treatment-specific changes in the response to ablative radiotherapy as opposed to ablative radiotherapy with concurrent ICB. Importantly, we nominate a genomic biomarker that predicts clinical benefit and improved survival following concurrent radiotherapy and ICB in metastatic NSCLC. These findings serve as a framework to guide the personalization of combinatorial radiotherapy and ICB strategies in patients with metastatic NSCLC and may be applicable to other human cancers.

Results

Cohort characteristics

A total of 37 patients with metastatic NSCLC were randomized to receive concurrent or sequential stereotactic body radiotherapy (SBRT) and ipilimumab plus nivolumab (ipi/nivo) immunotherapy as part of the randomized phase I trial to evaluate concurrent or sequential ipilimumab, nivolumab and stereotactic body radiotherapy in patients with stage IV non-small cell lung cancer (the COSINR study, [NCT03223155](https://doi.org/10.1038/s43018-022-00467-x))¹⁵. The CheckMate 227 trial previously demonstrated that ipi/nivo improved

clinical outcomes compared with nivo ($\pm$ platinum-doublet chemotherapy) or chemotherapy alone^{13,16}. With the future goal of determining whether SBRT further improves response rates to and survival following ipi/nivo, the COSINR phase I study evaluated the safety of combining multisite metastasis-directed SBRT to ipi/nivo in the first-line treatment of metastatic NSCLC. As recently reported, concurrent treatment resulted in fewer toxicities than sequential treatment¹⁵. In addition, concurrent treatment demonstrated a favorable objective response rate (ORR) and overall survival (OS) compared with the ipi/nivo arm of the CheckMate 227 trial (ORR, 44% versus 36%; 1-year OS, 84% versus 62%; 2-year OS, 62% versus 40%). The recently completed phase II component of the trial evaluated the efficacy of concurrent therapy in an expanded clinical cohort.

For all patients, pretreatment tumor biopsies were obtained prior to the administration of any therapy. Within the sequential arm, an on-treatment tumor biopsy was obtained after completion of SBRT, but prior to administration of ipi/nivo. Within the concurrent arm, an on-treatment biopsy was obtained after completion of SBRT and one cycle of ipi/nivo (Fig. 1a). The obtained pretreatment and on-treatment tumor biopsies were of the same irradiated metastatic lesion. Therefore, by comparing changes in matched tumor biopsies during treatment, we investigated the effect of SBRT versus SBRT + ipi/nivo on the tumor microenvironment (Fig. 1b). After review of pathologic and genomic data, 22 patients successfully underwent whole exome sequencing of matched tumor (pretreatment with or without on-treatment) and normal samples; total RNA sequencing (RNA-seq) of matched pretreatment and on-treatment tumor samples was successful for a subset of 15 patients (Extended Data Fig. 1a). A schematic of our analytical framework is shown in Extended Data Fig. 1b. Clinical and molecular data are available in Supplementary Tables 1–16.

Overall clinical and genomic characteristics of the cohort are presented in Extended Data Fig. 2. Clinicopathological characteristics were balanced between the concurrent and sequential treatment arms (Supplementary Table 17). A positive smoking history was associated with improved progression-free survival (PFS), whereas a greater number of disease sites and the presence of liver metastasis were associated with worse PFS; a larger number of disease sites was also associated with worse OS (Extended Data Fig. 3a). At a median follow-up of 17 months, there were no differences in PFS or OS between the treatment arms in the overall cohort (Extended Data Fig. 3b) or the subset of 22 patients in the molecular analysis (Extended Data Fig. 3c). In this context, we

investigated whether a distinct molecular subset of patients experienced differential outcomes following treatment.

Treatment-unique immunogenomic changes

We first characterized the differential genomic and transcriptomic effects of SBRT versus SBRT + ipi/nivo, which have not been previously reported in metastatic NSCLC. At the genomic level, we found substantial clonal evolution of somatic mutations after both SBRT and SBRT + ipi/nivo (Fig. 2a and Extended Data Fig. 4a). The majority of tumors treated with SBRT + ipi/nivo (5 out of 8) showed complete loss of tumor variants during therapy, suggesting complete or near-complete pathologic responses. The remaining tumors (3 out of 8) retained a clone with a core group of variants accompanied by at least one driver mutation alongside loss of existing and acquisition of new, somatic mutations. We investigated whether these genomic findings correlated with histologic reduction in the density of TTF1+/CK5+ cancer cells by performing multiplexed immunofluorescence analysis of matched formalin-fixed paraffin-embedded (FFPE) tumor specimens, which were independently obtained to validate the immunogenomic findings from fresh frozen tumor analysis. Tumors in which all genomic variants were eliminated on therapy had a significantly lower density of TTF1+/CK5+ cells than tumors that retained genomic variants on treatment (Wilcoxon $P = 0.03$) (Fig. 2b), whereas no differences were observed between groups at baseline ($P = 0.27$). By contrast, after SBRT, most tumors (8 out of 10) exhibited the latter phenotype of clonal persistence (Fig. 2a), suggesting less effective tumor cell elimination with SBRT compared with SBRT + ipi/nivo, as corroborated by multiplexed immunofluorescence analysis (Extended Data Fig. 4b). In addition, TMB and aneuploidy score, which is defined as the fraction of evaluable chromosomal arms that were amplified or deleted, were correlated at baseline (Spearman's $\rho = 0.54$, $P = 0.0095$) and decreased on treatment across both treatment arms (TMB median, 4.4 versus 1.2 mutations per megabase, paired Wilcoxon $P = 0.0013$; aneuploidy score median, 0.53 versus 0.04, paired Wilcoxon $P = 0.0015$); however, these results were primarily driven by changes following SBRT + ipi/nivo rather than SBRT alone (Fig. 2c). The reduction in TMB and aneuploidy score was also accompanied by a tumoral diploidization and reduction in purity (Extended Data Fig. 4c), suggesting that the loss of genomic variants during treatment was due to tumor cell elimination and subsequent enrichment of immune cells as opposed to a reduction of genomic instability within individual cancer cells. Similarly, clonal persistence in this setting reflected ineffective local treatment response.

Changes in gene expression were also dramatically different after SBRT compared with SBRT + ipi/nivo. SBRT + ipi/nivo upregulated, whereas SBRT decreased, immune signaling through IFN α , IFN γ , IL-6/JAK/STAT3 and inflammatory pathways (Fig. 3a and Extended Data Fig. 4d,e). In addition, SBRT + ipi/nivo, but not SBRT, decreased expression of G2/M cell cycle checkpoint, mitotic spindle, and E2F-dependent and MYC-dependent proliferation pathways (Extended Data Fig. 4d). We investigated whether these changes were related to an increase in stromal cells; however, neither SBRT nor SBRT + ipi/nivo increased the ESTIMATE (estimation of stromal and immune cells in malignant tumor tissues using expression data) stromal score (Extended Data Fig. 4f). In addition, there was no correlation between the change in single-sample

gene set enrichment analysis (ssGSEA) hallmark pathway scores and the change in ESTIMATE stromal scores during treatment, which suggested that treatment-induced changes in cell cycle and proliferation pathways following SBRT + ipi/nivo were due to reprogramming of remaining cancer cells rather than dilution with stromal cells.

In concert with these findings, there was an upregulation of immune genes involved in antigen presentation, interferon response, cytokine and chemokine signaling, and effector T cell function by SBRT + ipi/nivo (Fig. 3b). By contrast, SBRT suppressed expression of cytotoxic T cell genes. In addition, SBRT + ipi/nivo, but not SBRT, increased immune content as indicated by upregulation of xCell and ESTIMATE immune scores (Fig. 3c). Moreover, although analysis of immune cell signatures using xCell¹⁷ showed no baseline differences between treatment arms (Extended Data Fig. 5a), following treatment we observed a reduction in T cell expression signatures, particularly those representing CD8⁺ populations, after SBRT, whereas SBRT + ipi/nivo increased signatures of these key immune populations (Fig. 4a and Extended Data Fig. 5b).

The differential effect on effector T cells was further supported by intratumoral T cell receptor (TCR) analysis using *MiXCR*¹⁸. Overall, there were no changes in TCR richness or evenness in either treatment arm (Extended Data Fig. 5c); however, despite the lack of numerical changes in TCR diversity, individual TCRs showed a high degree of evolution. After SBRT, 641 out of 678 (94.5%) of TCRs present at baseline were eliminated on therapy, and 401 novel clonotypes were detected in the tumor. After SBRT + ipi/nivo, a similar portion of TCRs (380 out of 398, 95.5%) were eliminated on therapy; however, twofold more (764) novel clonotypes were present on therapy, suggesting that ICB facilitated the recruitment of a larger number of novel T cells into the tumor microenvironment (Fig. 4b and Extended Data Fig. 5d). Furthermore, TCR diversity correlated with the abundance of CD8⁺ T cells as represented by the xCell immune cell signatures (Extended Data Fig. 5e). Owing to the single sampling of TCRs in tumor samples and limited availability of fresh frozen tissue, we were unable to account for potential heterogeneity of TCR clonotypes across different tumoral regions^{19,20} or perform dedicated TCR sequencing. In addition, we further analyzed the cytolytic activity of the intratumoral T cells using an eight-gene effector T cell IFN γ -associated signature previously validated to predict ICB response in NSCLC¹⁴ and found a downregulation of the effector T cell signature after SBRT but an upregulation after SBRT + ipi/nivo (Fig. 4c). Importantly, although the T cell IFN γ -associated signature negatively correlated with FACETS tumor purity (Spearman's $\rho = -0.52$, $P = 0.008$), when changes in FACETS tumor purity were controlled for in a multivariable logistic regression model, we found that treatment arm was an independent predictor of change in the IFN γ -associated signature ($P = 0.000594$). This result indicated that the T cell IFN γ -associated signature was specifically upregulated following SBRT + ipi/nivo in the setting of overall changes in tumor purity.

The density of CD8⁺ T cells strongly correlated with both the expression of CD8⁺ T cell signatures (Extended Data Fig. 6a) and the expression of the T cell IFN γ -associated signature (Fig. 4d). This finding corroborated the result indicating that the transcriptomic signatures were reliable measures of activated, tumor-infiltrating CD8⁺ T cells. In the response to treatment, the change in CD8⁺ T cell density moderately

Fig. 2 | Therapy-induced changes in the genomic landscape. a, Clonal evolution of somatic mutations during treatment. Each box corresponds to an individual patient ($n = 18$ patients). Axes indicate the fraction of cancer nuclei (CCF) harboring the mutation; oncogenic mutations are highlighted in blue. All mutations with a calculable CCF (that is, FACETS purity > 0 and copy number available for the mutation locus) are plotted. **b**, Differences in on-treatment density of tumor (TTF1+/CK5+) cells as determined by multiplexed immunofluorescence between tumors in which no genomic variants were detected on therapy by whole exome sequencing (eliminated; patients 12, 13, 14, 32 and 49) and tumors with detected on-treatment variants (persistent)

(two-sided unpaired Wilcoxon test). Multiplexed immunofluorescence data are not available for patients 15 and 47 (SBRT, $n = 6$ patients; SBRT + ipi/nivo, $n = 6$ patients). The top and bottom edges represent the first and third quartiles, respectively. The center line represents the median. Whiskers extend to the furthest data points before any outliers (within $1.5 \times$ the interquartile range). Outliers are plotted as points above and below the box-and-whisker plot. **c**, Changes in aneuploidy score and TMB following treatment (two-sided paired Wilcoxon test; SBRT, $n = 10$ patients; SBRT + ipi/nivo, $n = 8$ patients). Box plot elements are the same as in part **b**.

with SBRT demonstrated a reduction in CD8⁺ T cell density (Fig. 4e and Extended Data Fig. 6c). Notably, intratumoral T cells showed a naive phenotype following SBRT + ipi/nivo as opposed to SBRT (Extended Data Fig. 6d). Collectively, these findings demonstrated that SBRT + ipi/



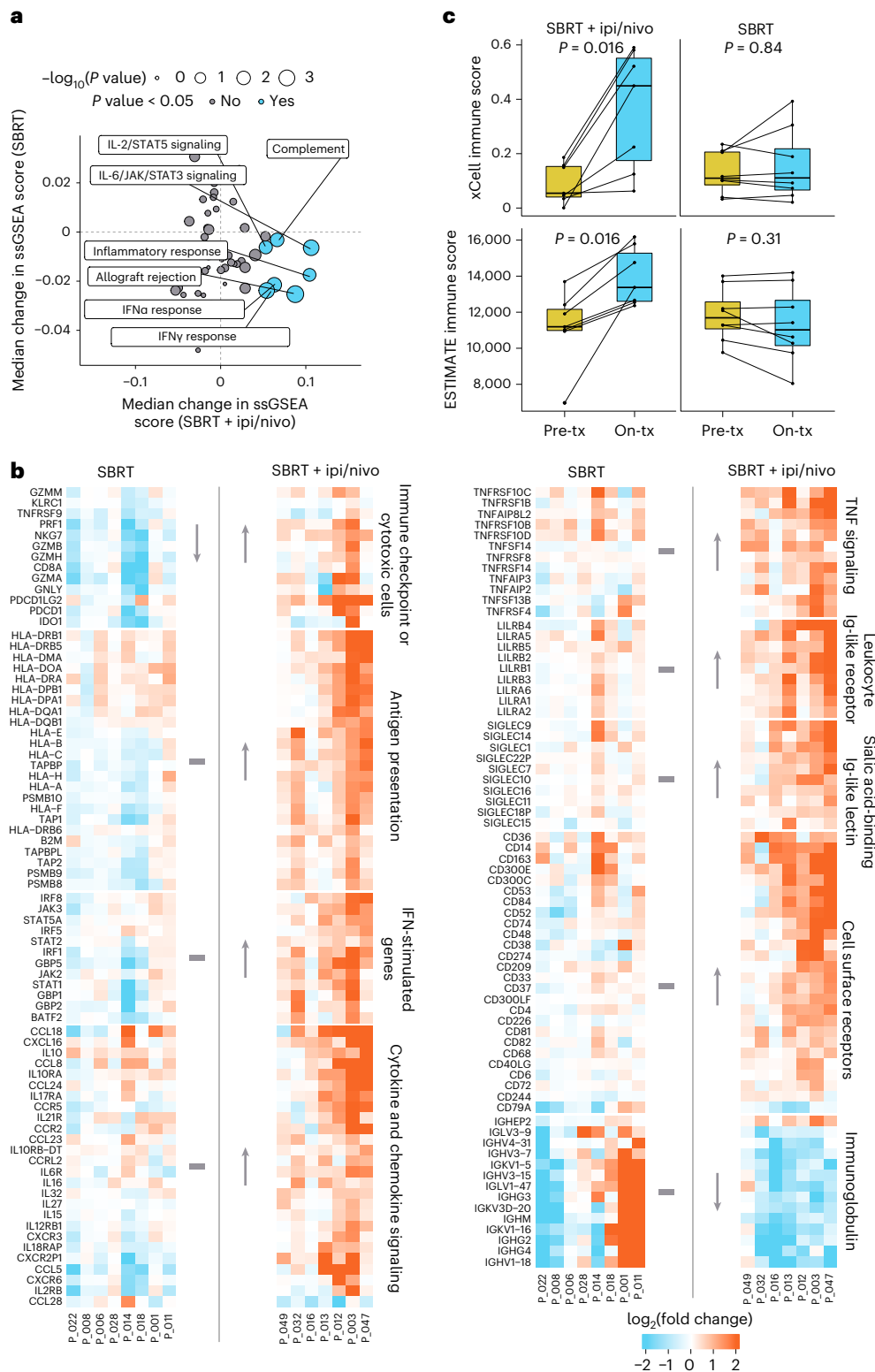


Fig. 3 | Therapy-induced changes in the tumor transcriptome. a, ssGSEA hallmark pathway changes during treatment in paired samples ($n = 15$ patients); two-sided Wilcoxon test for median changes (on-treatment (On-tx) – pretreatment (Pre-tx)) between treatment arms. Significant pathways are highlighted in blue ($P < 0.05$). **b**, Heatmap of fold changes ($\log_2(\text{on-treatment FPKM}/\text{pretreatment FPKM})$) in gene expression of manually selected immune genes found to be significantly upregulated or downregulated following SBRT

and/or SBRT + ipi/nivo ($\text{FDR} < 0.1$). Orange, upregulation; blue, downregulation. Upward and downward arrows reflect upregulation or downregulation of the genes in that cluster, respectively; dashed lines indicate no change. Columns represent patients (SBRT, $n = 8$ patients; SBRT + ipi/nivo, $n = 7$ patients). **c**, Changes in ESTIMATE and xCell immune scores in each treatment arm (two-sided paired Wilcoxon test; SBRT, $n = 8$ patients; SBRT + ipi/nivo, $n = 7$ patients). Box plot elements are the same as in Fig. 2b.

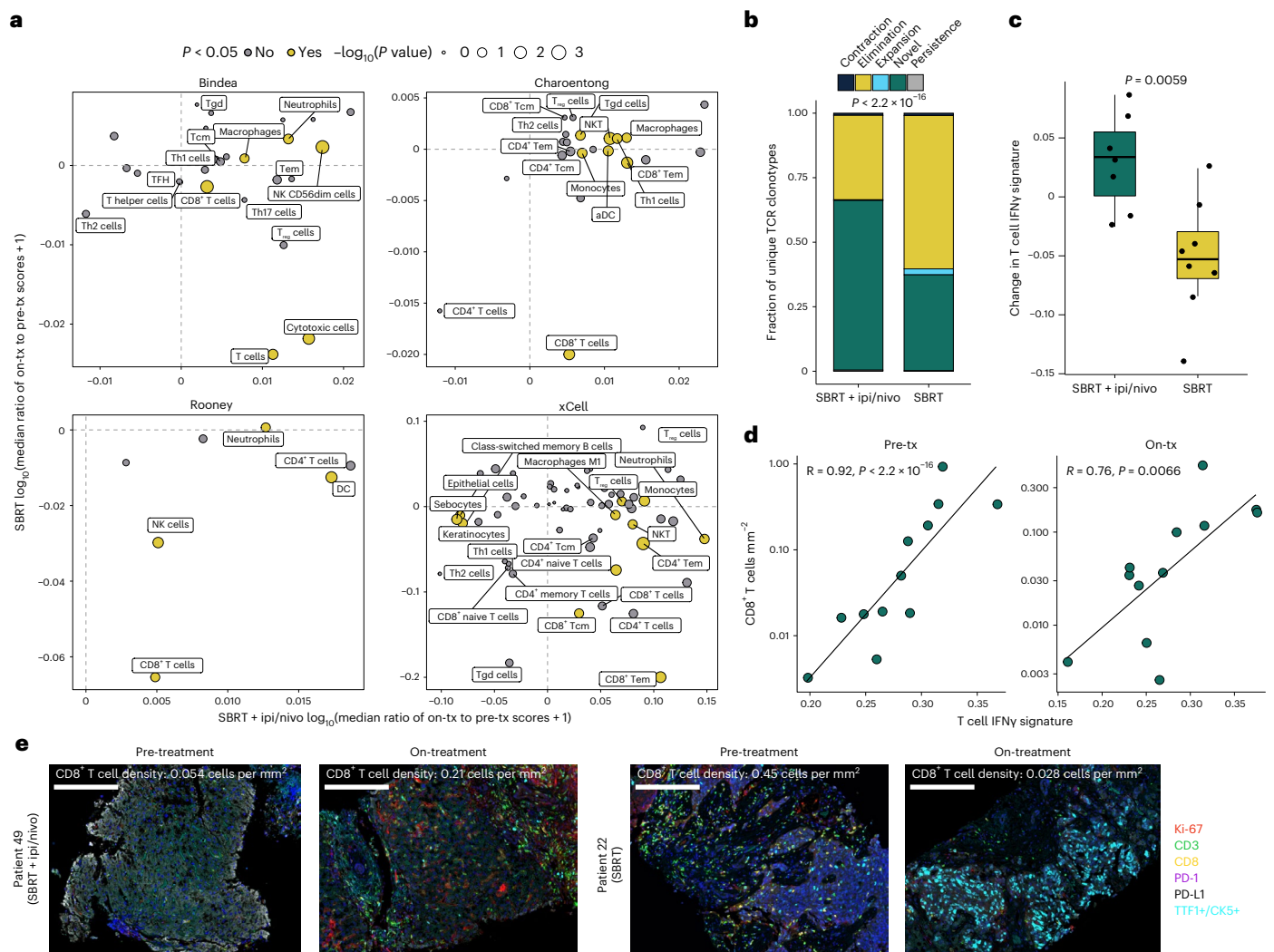


Fig. 4 | Immunological evolution during treatment. **a**, Differential changes in $\log_2$ (fold changes) in xCell immune cell-associated signatures following SBRT and SBRT + ipi/nivo. All T cell-associated signatures and any signature with $P < 0.05$ are labeled. Yellow points indicate $P < 0.05$. Two-sided Wilcoxon test comparing median $\log_2$ (fold change) of on-treatment versus pretreatment cell populations between the SBRT + ipi/nivo and SBRT groups (SBRT, $n = 8$ patients; SBRT + ipi/nivo, $n = 7$ patients). Significant (yellow) points are differentially changed cell populations following treatment, comparing the two treatment arms. **b**, Evolution of TCR clonotypes during treatment in paired samples (two-sided

Fisher's exact test; SBRT, $n = 8$ patients; SBRT + ipi/nivo, $n = 7$ patients). **c**, Change (on-treatment score – pretreatment score) in effector T cell IFN γ -associated gene signature (two-sided paired Wilcoxon test; SBRT, $n = 8$ patients; SBRT + ipi/nivo, $n = 7$ patients). Box plot elements are the same as in Fig. 2b. **d**, Correlation between IFN γ signature and CD8⁺ T cell density as determined by multiplexed immunofluorescence in pretreatment and on-treatment samples (two-sided Spearman correlation; SBRT, $n = 6$ patients; SBRT + ipi/nivo, $n = 6$ patients). **e**, Selected multiplexed immunofluorescence images from one tumor in each treatment arm. Scale bars, 50 μm ($\times 200$).

nivo more effectively eliminated tumor cells and amplified adaptive immunity compared with SBRT alone. In addition, these findings suggested that the stability in CD8⁺ T cell density following SBRT + ipi/nivo reflected a balance in the treatment-induced infiltration of activated CD8⁺ T cells and the loss of pre-existing tumor-resident CD8⁺ T cells.

Established immunotherapy biomarkers

Several biomarkers, including T cell IFN γ signature expression¹⁴, TMB, PD-L1 expression and neoantigen load, have been previously established as predictors of ICB response. Therefore, we examined whether these biomarkers were predictive of outcome in the context of SBRT + ipi/nivo. We found that none of these biomarkers were associated with PFS or OS, in the case of the entire cohort and in either treatment arm (Fig. 5a and Extended Data Fig. 7a–d). No specific mutations, gene-level copy number alterations (CNAs) or arm-level somatic CNAs (aSCNAs) were associated with PFS or OS, albeit our study was underpowered to detect such associations. In addition, we observed

no differences in the percentage of PD-L1-positive tumor or stromal cells at baseline, or in the response to treatment, between treatment arms (Extended Data Fig. 6e). Neither pretreatment nor change in the percentage of PD-L1-positive tumor or stromal cells was associated with survival across all patients or in either treatment arm. Therefore, we examined whether additional biomarkers could serve as predictors of survival following SBRT + ipi/nivo.

Recent studies suggest that highly aneuploid tumors show inferior responses to ICB, potentially due to mechanisms of immune evasion, such as downregulation of PD-L1 expression²¹ and reduction of tumor-infiltrating CD8⁺ T cells²². In our cohort, pretreatment aneuploidy score was differentially predictive of adverse outcomes in patients receiving sequential, but not concurrent, SBRT + ipi/nivo (OS, hazard ratio 94.9, 95% confidence interval 1.80–7633.4, $P = 0.04$; PFS, hazard ratio 74.4, 95% confidence interval 1.93–2922, $P = 0.021$) (Fig. 5a and Extended Data Fig. 7e). Aneuploidy score remained independently predictive of poor PFS ($P = 0.018$) and OS ($P = 0.08$) in patients receiving

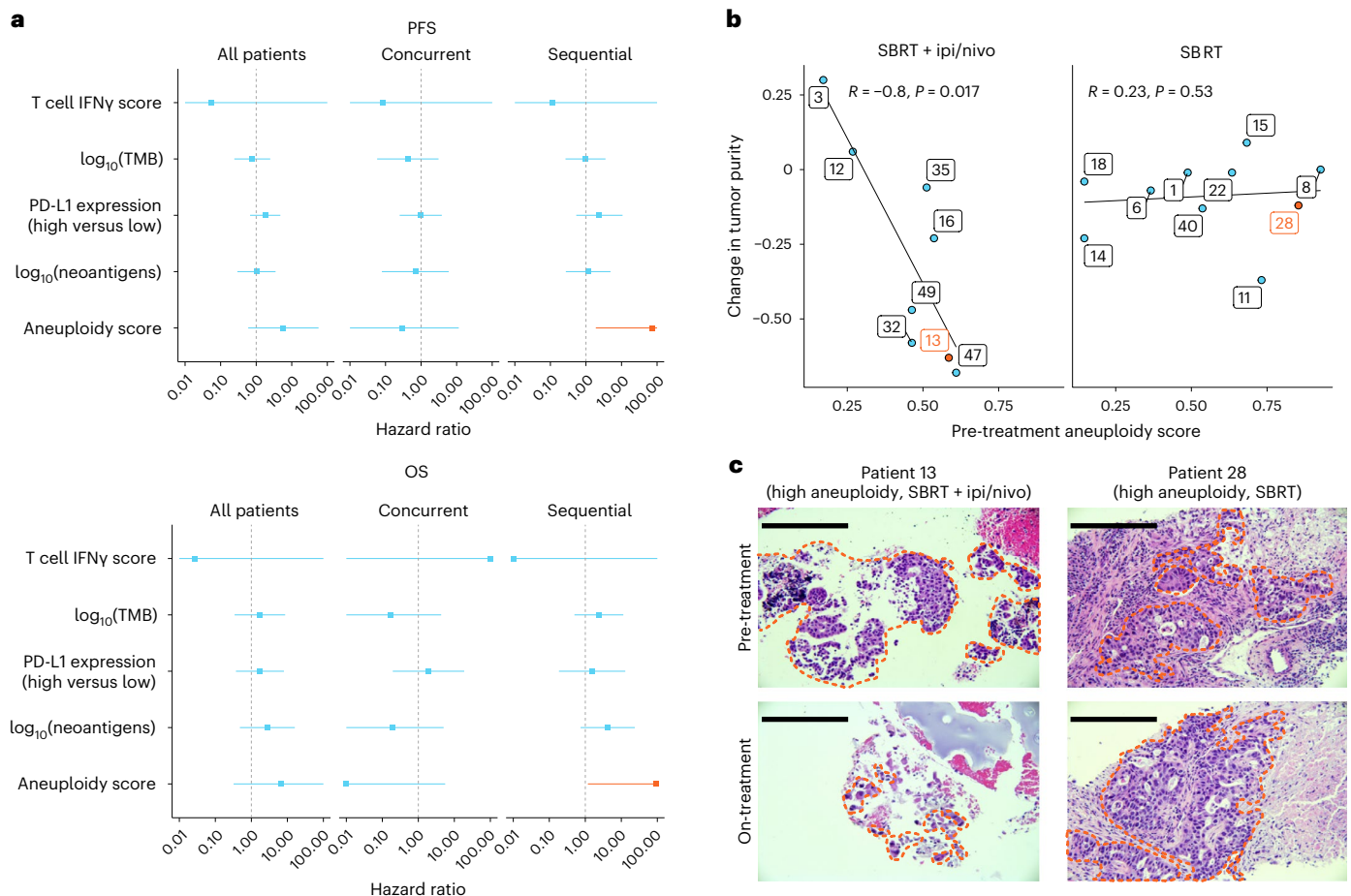


Fig. 5 | Evaluation of immunotherapy biomarkers. a, Forest plot demonstrating univariable hazard ratios for PFS and OS of established pretreatment biomarkers of immunotherapy response and aneuploidy score; PD-L1 expression (high ($\geq 50\%$) versus low ($< 50\%$) tumor proportion score (TPS)), $n = 34$ patients; TMB and aneuploidy score (continuous variables), $n = 22$ patients; neoantigens (continuous variable), $n = 18$ patients; and T cell IFN γ score (continuous variable), $n = 15$ patients. Center indicates hazard ratio point estimate, and lines indicate confidence intervals; confidence intervals exceeding limits of the x axis were truncated. **b**, Two-sided Spearman correlation between pretreatment aneuploidy score and change (on-treatment – pretreatment)

in tumor purity (SBRT, $n = 10$ patients; SBRT + ipi/nivo, $n = 8$ patients). Patient identification numbers are labeled, and those highlighted in orange are presented in part c. **c**, Representative images of pretreatment and on-treatment samples from high-aneuploid tumors treated with SBRT and SBRT + ipi/nivo (hematoxylin and eosin staining; scale bars, 50 μm ($\times 200$)). Both pretreatment samples show adenocarcinoma. The on-treatment sample from patient 13 predominantly shows necrosis with small areas of residual tumor compared with that from patient 28, which shows extensive residual adenocarcinoma alongside small regions of necrosis. Dotted orange lines outline viable tumor.

sequential therapy after controlling for TMB, as well as baseline prognostic clinical and pathological features (Extended Data Fig. 3a) (multivariable PFS (controlling for smoking history, number of disease sites and presence of liver metastases), $P = 0.04$; multivariable OS (controlling for number of disease sites), $P = 0.06$). These findings identified tumor aneuploidy as a potential biomarker of SBRT + ipi/nivo response in metastatic NSCLC.

We examined the relationship between baseline tumor aneuploidy and the change in tumor content following treatment, based on our previous findings that aneuploidy score and tumor purity decreased on-treatment. We found no correlation between baseline aneuploidy score and change in tumor purity after SBRT, although most samples demonstrated a small decrease in tumor purity following SBRT (Spearman's $\rho = 0.23$, $P = 0.53$) (Fig. 5b), probably secondary to tumor cell killing. By contrast, baseline aneuploidy score was highly negatively correlated with the change in tumor purity after SBRT + ipi/nivo (Spearman's $\rho = -0.80$, $P = 0.017$) (Fig. 5b), suggesting that highly aneuploid tumors were more effectively eliminated with SBRT + ipi/nivo than with SBRT. In support of these findings, pathologic review of tumor samples from highly aneuploid tumors demonstrated near-complete tumor responses following SBRT + ipi/nivo, but not following SBRT (Fig. 5c).

Aneuploidy predicts response to radiotherapy plus ICB in an independent cohort

Our finding that baseline aneuploidy was predictive of local tumor responses and survival led us to hypothesize that a pretreatment aneuploidy score could serve as a biomarker of the response to radiotherapy + ICB and survival. Aneuploidy score showed a moderate correlation with the FACETS fraction of genome altered (Spearman's $\rho = 0.51$, $P = 0.000724$), as aneuploidy score only encompasses arm-level changes, whereas the fraction of genome altered also includes focal events. Pretreatment aneuploidy score was negatively associated with immune signaling pathways, including IFN γ , TNF α , IL-2/STAT5 and inflammatory pathways (Fig. 6a) and ESTIMATE immune and xCell immune scores (Fig. 6b). In addition, patients with low PD-L1 expression, which is associated with reduced response to ICB, showed higher aneuploidy scores (Fig. 6c). Previous studies have demonstrated that aneuploidy is associated with *TP53* mutation, TMB and expression of proliferation genes²³. Notably, *TP53* mutation, TMB and expression of hallmark proliferation signatures (including mitotic spindle, G2/M checkpoint, E2F targets and MYC targets) were not associated with clinical outcomes in our cohort. In addition, aneuploidy score was not associated with other clinicopathologic features at baseline

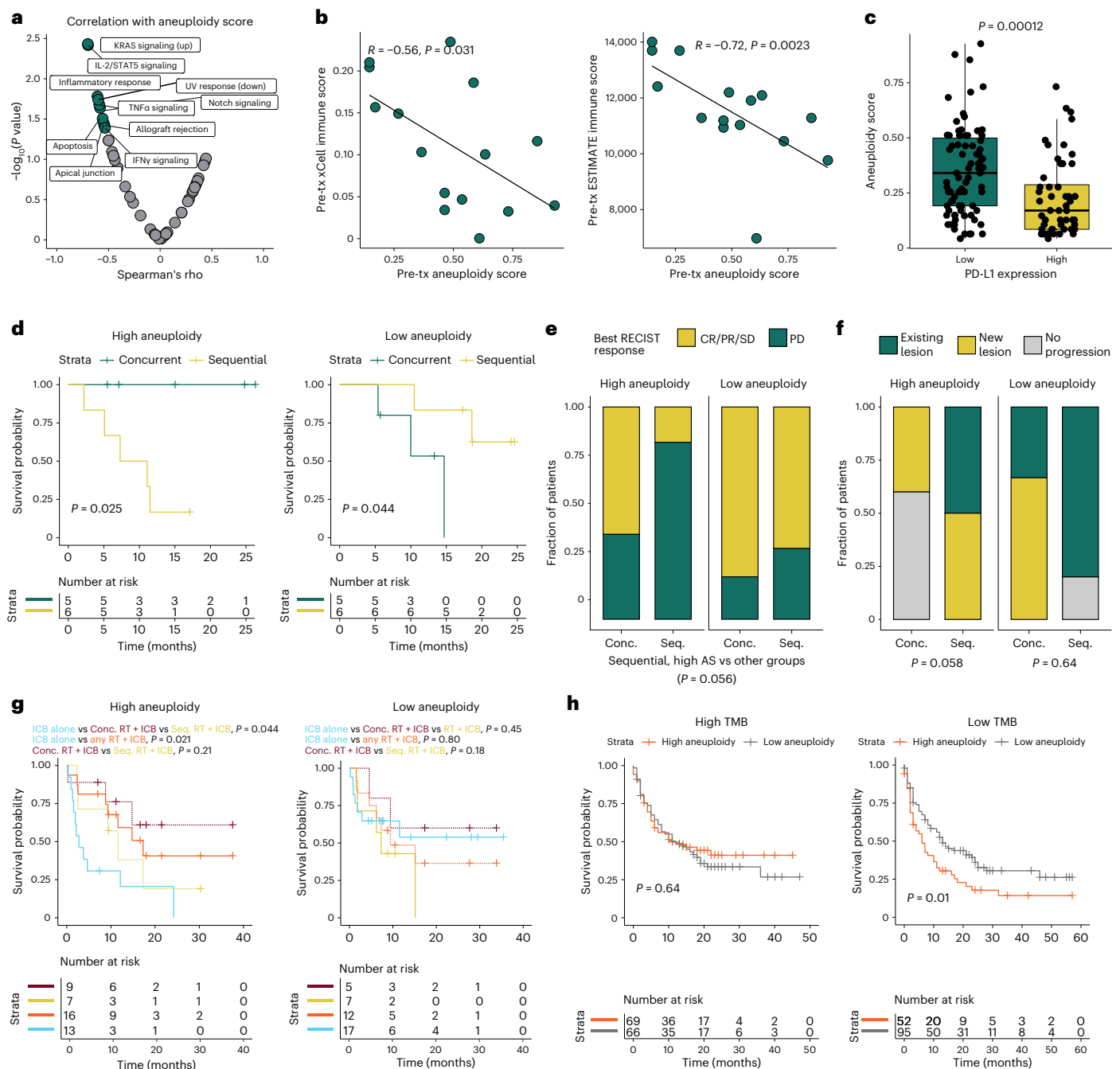


Fig. 6 | Anaploidy as a biomarker of combination radiotherapy and ICB response. a, Two-sided Spearman correlation between pretreatment aneuploidy score and ssGSEA hallmark pathways ($n = 15$ patients). Pathways highlighted in green are significant (two-sided $P < 0.05$). **b**, Correlation between pretreatment aneuploidy score, and ESTIMATE and xCell immune scores; two-sided Spearman correlation ($n = 15$ patients). **c**, Comparison of aneuploidy score by PD-L1 expression in the combined UC and COSINR cohorts (high PD-L1, $n = 52$ patients; low PD-L1, $n = 83$ patients); high PD-L1 is defined as $\geq 50\%$ TPS expression (two-sided Wilcoxon test). Box plot elements are the same as in Fig. 2b. **d**, Association between high aneuploidy score ($\geq$ median) and low aneuploidy score ($<$ median) and OS by treatment arm (log-rank test). **e**, Comparison of disease control rate (defined as lack of disease progression) in nonirradiated lesions by treatment arm and aneuploidy score (two-sided Fisher's exact test; for the concurrent (Conc.) arm, $n = 5$ patients for low aneuploidy score and $n = 5$ patients for high aneuploidy score; and for the sequential (Seq.) arm, $n = 6$ patients for low aneuploidy score

and $n = 6$ patients for high aneuploidy score). AS, aneuploidy score. **f**, Comparison of patterns of failure in nonirradiated lesions by treatment arm and aneuploidy score (two-sided Fisher's exact test). Yellow indicates the appearance of new lesions, and green indicates progression in an existing, unirradiated site. Sample sizes are defined in e. **g**, Association of treatment modalities with survival in high aneuploidy score ($\geq$ median) and low aneuploidy score ($<$ median) tumors in the UC cohort (log-rank test). Dotted lines represent subdivisions of the radiotherapy (RT) + ICB treatment group into patients treated with concurrent (maroon) or sequential (yellow) RT + ICB. **h**, Synergistic prediction of survival by TMB and aneuploidy score (tumors split by the cross-validated threshold identified in the UC cohort, ≥ 0.42 (high) versus < 0.42 (low)) in an independent metastatic NSCLC cohort treated with immunotherapy. Orange lines, patients with high aneuploidy score (≥ 0.42); gray lines, patients with low aneuploidy score (< 0.42) (two-sided log-rank test). CR, complete response; PD, progressive disease; PR, partial response; SD, stable disease.

(Supplementary Table 18). However, the lack of association between specific genomic alterations or proliferation signatures and survival was likely to have been limited by the modest sample size of our study; therefore, we cannot conclude that aneuploidy score is independently associated with clinical outcomes.

We further examined the relationship between aneuploidy score and survival. We found that patients whose tumors exhibited high aneuploidy (aneuploidy score $\geq$ cohort median, 0.54; interquartile range, 0.32–0.78) had improved survival when treated with concurrent as opposed to sequential SBRT + ipi/nivo (12-month OS, 100% versus 17%, log-rank $P = 0.025$) (Fig. 6d). PFS followed similar trends to OS (Extended Data Fig. 8a). By contrast, this relationship was not observed in patients with less aneuploid tumors (12-month PFS, 20% versus 17% (concurrent versus sequential), log-rank $P = 0.44$; 12-month OS, 53% versus 83% (concurrent versus sequential), log-rank $P = 0.044$) (Fig. 6d). Although there were three deaths in the concurrent arm and two deaths in the sequential arm for low-aneuploidy tumors, one death in the concurrent arm was that of a patient for whom there was no evidence of active cancer. Instead, the death was attributed to baseline pulmonary disease. All other deaths of patients included in this study were preceded by disease progression. When we examined cancer-specific survival for low-aneuploidy tumors, there was no difference between the two treatment arms (log-rank $P = 0.24$).

We further examined the treatment-related effects on nonirradiated tumor sites. Within the subset of high-aneuploidy tumors, distant tumor responses in nonirradiated lesions occurred in 17% of patients treated with sequential therapy and 60% of patients treated with concurrent therapy, which was similar to the percentages seen in low aneuploidy tumors irrespective of treatment arm (Fig. 6e). In addition, high-aneuploidy tumors treated with concurrent therapy were less likely to experience disease progression in an existing unirradiated site compared with tumors treated with sequential therapy (Fig. 6f). Importantly, the associations between high aneuploidy score and decreased survival were not confounded by baseline tumor purity or the number of disease sites (Extended Data Fig. 8b,c). These findings suggested that in highly aneuploid tumors, concurrent therapy increased abscopal responses in nonirradiated lesions compared with sequential therapy, and improved survival.

We examined whether the effect of tumor aneuploidy on the differential responses to radiotherapy and ICB could be validated in independent clinical metastatic NSCLC cohorts. We reasoned that concurrent radiotherapy and ICB benefits patients whose tumors demonstrate high aneuploidy due to their adverse baseline prognosis. To test this hypothesis, we investigated an independent contemporary clinical cohort of 58 patients with metastatic NSCLC who underwent next-generation targeted genomic sequencing prior to ICB initiation at our institution (The University of Chicago (UC) cohort)²⁴ (Methods). Patients received either ICB alone (anti-PD-1 or anti-PD-L1 inhibitor $\pm$ cytotoxic chemotherapy) or ICB with radiotherapy to at least one extracranial disease site during (concurrent) or preceding or following (sequential) ICB. The compositions of clinical and pathological factors were balanced across the COSINR and UC cohorts (Supplementary Table 19). Consistent with our hypothesis, we found that after ICB alone, high aneuploidy score (aneuploidy score $\geq$ cohort median, 0.40; interquartile range, 0.21–0.53) was associated with reduced tumor responses (complete response, partial response or stable disease, 10% versus 50% (high versus low aneuploidy score), Fisher's exact $P = 0.07$) and OS compared with low aneuploidy score (12-month OS, 33% versus 56% (high aneuploidy score versus low aneuploidy score), log-rank $P = 0.06$). Importantly, patients with highly aneuploid tumors who received radiotherapy + ICB had higher 12-month survival than those who received ICB alone (59% versus 31%, log-rank $P = 0.021$) (Fig. 6g). Moreover, among the patients who received radiotherapy + ICB, those who received the two modalities concurrently experienced better outcomes than those who received them sequentially, although this

did not reach statistical significance (12-month OS, 76% versus 38%, $P = 0.21$). By contrast, no survival improvement was observed in adding radiotherapy to ICB in patients with low aneuploidy score tumors (Fig. 6g). Among known prognostic factors, worse Eastern Cooperative Oncology Group (ECOG) performance status (2 or 3 versus 0 or 1), a greater number of disease sites and the presence of liver metastases were associated with adverse OS in the UC cohort (Extended Data Fig. 8d). However, in a multivariable model that includes these variables, the addition of radiotherapy to ICB (versus ICB alone) remained independently predictive of improved OS in patients with highly aneuploid tumors (aneuploidy score $\geq$ cohort median, Wald $P = 0.016$).

To identify an optimal threshold to define high aneuploidy, we performed a leave-one-out cross-validation analysis comparing data from patients who received ICB alone with that of patients who received radiotherapy + ICB in the UC cohort. We defined the optimal aneuploidy score threshold in the UC cohort and not in the COSINR dataset because the distribution of aneuploidy scores was more right skewed in the COSINR cohort than in the UC cohort and in a third cohort of 500 lung adenocarcinomas from The Cancer Genome Atlas (TCGA)²⁵ (UC versus TCGA Kolmogorov–Smirnov $P = 0.26$; UC or TCGA versus COSINR $P = 0.025$) (Extended Data Fig. 8e). Therefore, using the UC cohort allowed us to maximize generalizability of an optimal threshold to other datasets. We determined that an aneuploidy score threshold of 0.42 represented the optimal cutoff in the UC cohort (6-month OS, 85% versus 31% (radiotherapy + ICB versus ICB only), log-rank $P = 0.014$) (Extended Data Fig. 8f,g). Importantly, applying this cross-validated aneuploidy score threshold to the COSINR cohort also demonstrated promise as a threshold for high aneuploidy score (12-month OS, 88% versus 33% (concurrent arm versus sequential arm), log-rank $P = 0.09$) (Extended Data Fig. 8h).

Last, we examined the relationship between aneuploidy score, TMB and survival in an independent cohort of 350 patients with metastatic NSCLC who received ICB at the Memorial Sloan Kettering Cancer Center (MSKCC)¹⁰. In a multivariable Cox proportional hazards model, TMB and aneuploidy score independently predicted OS (TMB, hazard ratio 0.96, Wald $P = 0.00016$; aneuploidy score, hazard ratio 2.25, $P = 0.014$ (per unit increase)). We proposed that a high aneuploidy score could independently identify a subset of patients with metastatic NSCLC with an adverse prognosis and split patients by the median TMB to compare the OS of patients whose tumors exhibited a high aneuploidy score (≥ 0.42) with those whose tumors exhibited a low aneuploidy score (< 0.42). The data demonstrated that among low TMB tumors, high aneuploidy score tumors showed significantly worse OS (12-month OS, 30% versus 52% (high aneuploidy score versus low aneuploidy score), log-rank $P = 0.01$), further supporting the utility of our high aneuploidy score threshold as a biomarker of ICB response in metastatic NSCLC (Fig. 6h). Taken together, our findings suggest that high aneuploidy score metastatic NSCLCs show a deleterious baseline prognosis with ICB alone but derive the greatest clinical benefit from the concurrent use of radiotherapy and ICB.

Discussion

Our findings challenge the prevailing view that ablative radiotherapy induces a favorable intratumoral adaptive immune response, reveal differential immunogenomic consequences of ablative radiotherapy versus concomitant radiotherapy and ICB, and establish high tumor aneuploidy as a potential biomarker of radiotherapy and ICB response. By comprehensively analyzing the molecular features of matched pre-treatment and on-treatment tumor biopsies from patients treated on a randomized clinical trial, we determined that SBRT combined with ipi/nivo increased the expression of adaptive immune and cytotoxic T cell gene programs, increased TCR clonotypic diversity and improved tumor cell elimination over SBRT alone. These findings support preclinical studies that demonstrated upregulation of major histocompatibility complex class I expression, augmentation of immunogenic cell death

and cytokine expression, and improvements in CD8⁺ effector T cell function with combination radiotherapy and ICB^{2-7,26}. However, our findings demonstrated that SBRT alone was insufficient in metastatic NSCLC to induce local immune augmentation that has been suggested in previous studies^{2,5,27}. The lack of induced innate and adaptive immunity by SBRT in our study may be due to the timing of the on-treatment biopsy. Most preclinical studies have measured cytokine induction and CD8⁺ T cell priming at relatively short intervals following radiotherapy (for example, 10–14 days), and the apparent lack of a positive immune response observed here (median 27.5 days between biopsies) may be related to the durability of immunity induced by radiotherapy alone rather than lack of immune activation by radiotherapy.

We identified elevated tumor aneuploidy score as a predictor of survival following radiotherapy and ICB in metastatic NSCLC, which supports recent findings in patients with metastatic melanoma treated with ICB²² and NSCLC treated with radiotherapy²⁸. Extending these findings, we demonstrated that radiotherapy to extracranial disease sites concomitant with, but not before or after, ICB improves the adverse baseline prognosis of patients with highly aneuploid tumors, which we suggest is due to augmentation of local and distant tumor immunity. By contrast, no survival benefit was detected with the addition of radiotherapy to ICB in patients with less aneuploid tumors. We propose that tumors exhibiting elevated aneuploidy derive the greatest benefit from the addition of concurrent radiotherapy and ICB because concurrent therapy elicits a more rapid and deeper local tumor response (that is, greater clonal elimination combined with increased immune infiltration) compared with sequential therapy, which ultimately affects systemic disease response and survival. On a practical level, aneuploidy can be readily obtained from targeted genomic sequencing panel data using existing methods²⁹, as demonstrated in our UC validation cohort. Future trials of radiotherapy and ICB are needed to validate aneuploidy as a biomarker.

Although further studies are needed to determine a mechanism for the observed disparity in outcomes among highly aneuploid tumors treated with sequential versus concurrent radiotherapy and ICB, it is possible that this is driven in part by the previously described inherent resistance of highly aneuploid tumors to radiotherapy²⁸. Aneuploidy may also induce immune suppression through induction of proteotoxic stress due to an increase in gene products created by arm-level SCNAs as well as less effective neoantigen major histocompatibility complex binding and presentation. However, the addition of ICB to radiotherapy may partially overcome these immunosuppressive effects by improving antigen processing and presentation while recruiting novel CD8⁺ T cells into the tumor microenvironment. Concomitantly, immunogenic cell death due to SBRT could create damage-associated molecular patterns that can then be recognized by the novel T cell repertoire. Administration of ICB after radiotherapy as opposed to simultaneously may limit the beneficial effect of the addition of radiotherapy.

Despite the limited sample size of matched metastasis biopsies that passed our stringent quality standards, this is nevertheless a comparatively large biopsy-driven study of SBRT in NSCLC and examines the impact of radiotherapy versus radiotherapy + ICB. This allowed us to develop and test hypotheses related to tumor aneuploidy in two independent metastatic NSCLC cohorts. However, an important caveat of our analysis is that the on-treatment biopsy in the sequential arm does not reflect the post-ICB response. Thus, it is possible that the immediate immunosuppressive effects following SBRT can be attenuated by the sequential administration of ICB, as recently suggested³⁰. In addition, the limited quantity of tumor tissue, as well as the moderate genomic sequencing depth, constrained our ability to assess intratumoral heterogeneity of treatment-induced clonal evolution. A fraction of tumors in our study exhibited low tumor purity, which can affect copy number identification as well. Nevertheless, among highly aneuploid tumors, we found improved survival with concurrent as opposed to sequential therapy, suggesting that the timing of radiotherapy and ICB, and the

resulting immunogenomic consequences contribute to the survival differences among patients with metastatic NSCLC. The identification of aneuploidy as a biomarker of benefit of concurrent radiotherapy and ICB in two independent metastatic NSCLC cohorts suggests that this observation is reproducible.

Taken together, our results represent a comprehensive immunogenomic study of the effects of combination SBRT and ICB in metastatic NSCLC. We demonstrate the immunogenic interaction of concurrent SBRT and ICB, and identify tumor aneuploidy as a potential biomarker of the response to combination radiotherapy and ICB. Our findings suggest that existing clinical genomic sequencing architecture can inform strategies to optimize the integration of radiotherapy and ICB currently being tested in clinical trials of metastatic NSCLC and many other human cancers.

Methods

Patient selection and sample collection

This study complies with all ethical regulations and was approved by the University of Chicago Biological Sciences Division Institutional Review Board Committee (IRB17-0547-CR003). Patients enrolled in the COSINR study¹⁵ (NCT03223155) were eligible for analysis. The COSINR study is a randomized phase I/II trial designed to evaluate the safety and efficacy of combination ICB using ipi/nivo plus sequential or concurrent SBRT as a first-line treatment for patients with stage IV NSCLC. Eligibility criteria include ECOG 0 or 1 performance status, measurable disease defined by Response Evaluation Criteria in Solid Tumours (RECIST) v1.1, no symptomatic central nervous system disease and no autoimmune or immunodeficiency syndromes. Without consideration of PD-L1 expression or TMB, patients were randomized to SBRT to two to four metastatic sites with concurrent or sequential (within 7 days) immunotherapy. A total of 37 patients were treated on the phase I component of the COSINR study. Matched tumor biopsies of a single irradiated metastasis were obtained prior to treatment and following completion of SBRT (sequential arm) or completion of SBRT and one cycle of ICB (concurrent arm). Biopsies were obtained approximately 3 to 4 weeks apart. PD-L1 expression was determined as the percentage of tumor cells expressing PD-L1 by clinical immunohistochemistry testing, as previously described¹⁵. Diagnostic imaging with positron emission tomography–computed tomography and whole-body computed tomography was performed with cycle 2 and computed tomography was at minimum performed with every other cycle thereafter until treatment completion or discontinuation (Fig. 1a). Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Of 37 eligible patients, 22 had successful whole exome sequencing of pretreatment biopsies as well as matched peripheral blood; of these, on-treatment biopsies for 18 patients were included. Of the 18 patients, 15 underwent successful pretreatment and on-treatment RNA-seq and were included in our analysis (Consolidated Standards of Reporting Trials (CONSORT) diagram shown in Extended Data Fig. 1b). Although additional fresh frozen tumor samples were unavailable as all of these samples were used for whole exome sequencing and RNA-seq, we previously acquired independent FFPE core biopsies from each patient's irradiated metastasis at the specified time points, which allowed us to assess the concordance between the fresh frozen and FFPE biopsies. Manual review of pathologic and sequencing data (mutations and copy number profiles) was conducted to ensure that tumor variants were detected in all samples. We identified two patients (12 and 13) who underwent biopsy and who had subsequent molecular analysis of a partially irradiated metastasis, which was permitted in the study. In both patients, the planning target volume covered 95–99% of the original internal target volume; therefore, the majority of the gross tumor received the full radiation prescription dose. As such, we included patients 12 and 13 in the molecular analysis. Patients with diploid tumors or those with no mutations in pretreatment samples were

excluded. In addition, on-treatment samples in which no tumor variants were identified were assigned diploid status (ploidy = 2, purity = 0). These samples were used only to demonstrate elimination of tumor variants and for analysis of local immune transcriptomic milieu if immune and stromal elements were detected on pathologic review.

Validation of our findings was performed using a 58 patient subset of 139 patients with NSCLC who underwent next-generation targeted genomic sequencing and were treated with ICB (59% anti-PD-1 or anti-PD-L1 and 41% ICB in combination) at our institution, as previously described²⁴. We selected patients with available segmented copy number data and split those patients into three well-defined subsets for further analysis: (1) ICB alone (no radiation; patients who received radiotherapy to the brain (radiosurgery or whole-brain radiation therapy) were excluded, $n = 30$); (2) sequential radiotherapy and ICB (interval between treatments of 7–60 days; $n = 14$); and (3) concurrent radiotherapy and ICB (interval between treatments of 0 days; $n = 14$). In the sequential and concurrent groups, patients were selected only if they had received radiotherapy to at least one extracranial disease site regardless of radiotherapy dose or target organ site.

Copy number segmentation files for 500 lung adenocarcinoma samples from the TCGA Pan Cancer Atlas dataset²⁵ were downloaded from cBioPortal³¹ for analysis of the distribution of aneuploidy scores in a large multi-institutional cohort. In addition, copy number and clinical data for a cohort of 350 NSCLCs treated with ICB (94% anti-PD-1 or anti-PD-L1 and 6% ICB in combination) at the MSKCC¹⁰ were downloaded from cBioPortal to examine the relationship between TMB, aneuploidy and survival in the setting of ICB.

DNA and RNA extraction and sequencing

Tumor DNA and RNA samples were isolated from snap-frozen tissue biopsies using the Qiagen Allprep DNA/RNA Mini kit according to the manufacturer's instructions. Germline DNA samples were isolated from the whole blood using the Qiagen PAXgene Blood DNA kit according to the manufacturer's instructions. Tumor RNA samples were treated with DNase, and quality control was performed using the Agilent RNA 6000 Pico kit. Ribo-Zero total RNA libraries were prepared and sequenced on the Illumina NovaSeq 6000 system at a depth of approximately 60 million reads per sample. Tumor DNA and germline samples were treated with RNase, and quality control was performed using the Agilent High Sensitivity DNA kit and Agilent Genomic DNA Fragment Analyzer. SureSelect V7 libraries were then prepared and sequenced on the Illumina NovaSeq 6000 system.

Whole exome sequencing sample processing and variant calling

Raw fastq files were first trimmed using Trimmomatic v0.39³². The trimmed reads were aligned to the hg38 human reference genome³³ using BWA-mem v0.7.1³⁴ and sorted using samtools v1.11³⁵. PCR duplicates were identified using Picardtools v2.23.8 MarkDuplicates and further recalibrated using GATK³⁶ v4.1.9.0 BaseRecalibrator and ApplyBQSR with known indels from the GATK resource bundle in the Agilent sureSelect Human exome V7 bait set. Somatic variant calling for each tumor–normal pair was performed using GATK4 Mutect2. The called somatic variants were further filtered by GATK FilterMutectCalls using contamination estimates from CalculateContamination. The filtered VCF was annotated with ANNOVAR v20191024³⁷ and GATK funcotator to generate a MAF file for each tumor sample. The variants underwent 8-oxoguanine (8-oxoG) filtering, as described below. Further filtering was performed wherein variants with a variant allele fraction of ≤ 0.1 or with ≤ 5 supporting variant reads were excluded to reduce the burden of sequencing artifacts. We rescued variants that were listed as pathogenic or likely to be pathogenic in OncoKB (described below), were seen in COSMIC v91 (database downloaded on June 30, 2021) at least three times or were called in the paired sample when both pretreatment and on-treatment samples were available. FACETS³⁸ v0.6 was used to

call somatic CNAs and determine sample purity and ploidy using the default settings.

8-oxoG artifact filtering

Owing to the observed high burden of C > A transversions at low variant allele fractions, we determined that it was necessary to perform additional filtering to remove 8-oxoG sequencing artifacts. We performed this filtering, as previously described³⁹. In brief, C > A or G > T mutations were interrogated for a potential artifactual origin by computing the fraction of 8-oxoG (FoxoG) as the reads with G > T changes on read 1 and C > A changes on read 2 based on the strand-specific read counts as determined using MuTect2:

$$\text{FoxoG} = \begin{cases} \text{if } C > A : F2R1/(F1R2 + F2R1) \\ \text{if } G > T : F1R2/(F1R2 + F2R1) \end{cases}$$

We then removed all mutations where

$$\text{Tumor_LOD} < -10 + (100/3) \times \text{FoxoG}.$$

Determination of pathogenicity of genomic alterations

Predicted mutation oncogenicity was retrieved from the OncoKB⁴⁰ application programming interface (v3.1) using the peptide change on September 21, 2021. Mutations annotated as oncogenic or likely/predicted oncogenic were considered pathogenic. For gene-level copy number variants, we included only those afflicting known oncogenes or tumor suppressors as defined in the OncoKB database. We defined amplifications as loci (genes) called by FACETS as having a total copy number (TCN) $\geq$ (the average sample ploidy + 3) and homozygous deletions as those with a TCN = 0.

CCF calculation

Cancer cell fractions (CCFs) were obtained for each mutation using previously described methods⁴¹. The expected variant allele fraction (VAF) was calculated for each potential CCF ($\widehat{CCF}$) between 0.001 and 1 in increments of 0.001 using the FACETS sample purity (p), allelic copy number (ACN) and TCN at the mutation locus.

$$E(\text{VAF}) = \frac{p \times \widehat{CCF} \times \text{ACN}}{2 \times (1 - p) + p \times \text{TCN}}$$

The final CCF of the mutation was the candidate CCF that maximizes the binomial probability of observing the alternate (n_{alt}) and reference (n_{ref}) reads at the mutant locus.

$$\text{CCF} = \widehat{CCF} \arg\max [\text{Binom}(n_{\text{alt}} | n_{\text{ref}}, E(\text{VAF}))]$$

TMB and aneuploidy score calculation

TMB was defined as the number of nonsynonymous mutations in each sample divided by the sequencing bait size (COSINR, 35.7 Mb; UC, 2.8 Mb). Arm-level SCNAs were called using ASCETS²⁹ version 1.1 on the normalized copy number segmentation files from FACETS (COSINR) or CNVkit (UC), using a threshold for calling amplifications and deletions of ± 0.1 . The aneuploidy score²³ for a sample was defined as the fraction of evaluable arms (ASCETS call of AMP, DEL, NEUTRAL or NC) afflicted by arm-level SCNAs (AMP or DEL). ASCETS was specifically designed and validated for use in targeted panel data and was therefore used in the UC cohort, as previously described²⁹.

Mutational signature deconvolution

Mutational signature deconvolution was performed using the MutationalPatterns⁴² package (v4.1.1) in R on all samples with at least 20 mutations ($n = 32$). After excluding mutations possibly to represent 8-oxoG artifacts, we performed an initial fit of the COSMIC signatures to the refined mutation spectra in our sample using the

fit_to_signatures_strict() function with a max_delta parameter of 0.02. We observed examples of signature overfitting despite the strict refitting approach, with a multitude of mutational processes predicted to be active in each sample, including many that were not expected to be found in NSCLC. Therefore, we refit the spectra using the list of only those signatures demonstrated to be found in lung adenocarcinoma or lung squamous tumors as previously described⁴³ and using information on the COSMIC website.

To confirm that we had successfully removed artifactual mutations during 8-oxoG filtering, as described above, we compared the mutation spectrum of the filtered mutations to the COSMIC signatures. As expected, the most similar COSMIC signature was SBS45 (oxidative damage by 8-oxoG) with *cossim* = 0.96.

RNA-seq pipeline

RNA-seq analysis was carried out by first aligning 100-bp-length paired-end reads to the human hg38 genome using the STAR⁴⁴ aligner v2.6.1d. The resulting BAM files were sorted by read name using samtools³⁵ v1.10. The total reads per gene were counted using htseq with the stranded option and using the Ensembl human hg38 list of coding exons for each gene as a reference. A matrix table of counts for each gene and sample was generated. The resulting table was analyzed for differential expression as paired samples (pretreatment versus on-treatment) using the R package DESeq2⁴⁵ using count values. Log₂(fold change) values were shrunk using the apeglm method⁴⁶.

Single-sample gene set enrichment analysis

ssGSEA was performed using the *gsva* package in R on the normalized RNA-seq FPKM (fragments per kilobase per million reads) matrix with the following parameters (method = 'ssgsea', kcdf = 'Poisson') on the set of hallmark gene sets (h.all.v7.4.symbols.gmt).

Immune signature analysis

Cell-specific immune signatures were calculated using the xCell¹⁷ web portal (<https://xcell.ucsf.edu/>) using all signature matrices that included T cells (xCell, Bindea, Charoentong and Rooney) on the normalized expression (FPKM) matrix of all 30 paired RNA-seq samples. Effector T cell IFN γ ¹⁴ and naive T cell signatures⁴⁷ were calculated using the R package *singscore*.

HLA typing and neoantigen binding prediction

Computational human leukocyte antigen (HLA) typing was performed on each matched normal sample using PolySolver v4.0⁴⁸. For the neoantigen and neopeptide analyses, we performed a modified version of the approach described in⁴⁹. Eight amino acids on either side of each nonsilent single-nucleotide variant with an annotated protein change were retrieved using the biomaRt⁵⁰ package, forming a 17-mer (17-base polymer) peptide. This peptide was transformed into 9-mer peptides using a sliding window approach. We then used NetMHCpan⁵¹ v4.1 to predict the binding of each candidate neopeptide to the patient-specific HLA alleles. Neopeptides with a rank <2% were considered to be binders. We then excluded any neopeptides that were not expressed in the harboring sample (FPKM = 0). This enabled the analysis to at least partially capture the effects of upregulating and downregulating expression of neoantigens in on-treatment samples. Although an individual mutation could potentially create multiple neopeptides, in our analysis, *neoantigens* specifically refer to a subset of neopeptides in which each mutation can represent at most one neoantigen.

TCR clonotype analysis

TCR clonotypes were determined from bulk RNA-seq using the MiXCR¹⁸ pipeline. The resulting clonotypes were used to calculate richness (the number of unique clonotypes in a sample) and evenness (the Shannon entropy of the clonotypes in a sample divided by the maximum possible entropy given the number of clonotypes (equal to log₂(richness))).

Changes in TCR populations on therapy were categorized as follows: (1) novel (new clonotypes detected only in on-treatment sample); (2) elimination (clonotypes in pretreatment samples not detected on therapy); (3) expansion (existing clonotypes whose clone fraction increased by at least 10% on therapy); (4) contraction (existing clonotypes whose clone fraction was reduced by at least 10% on therapy); and (5) persistence (existing clonotypes whose clone fraction changed less than 10% on therapy).

Treatment response and survival analysis

RECIST v1.1⁵² was used to assess unirradiated tumor response. Irradiated tumors were not evaluated as part of the treatment response, consistent with RECIST principles. PFS was defined as the time from starting RT to progression or death, and OS as the time from starting RT to death. Patients were scored as censored at the time of the last follow-up. Survival analysis was performed using the *survminer* and *survival* packages, using the survdiff() function in R for Kaplan–Meier survival analyses. For univariable and multivariable Cox proportional hazards analysis, the coxph() function was used. Unadjusted or adjusted hazard ratios for survival were reported as appropriate. For analyses of specific genomic predictors of response, we included as covariates TMB for mutations and aneuploidy score for CNAs to correct for background levels of genomic instability.

Leave-one-out cross validation

Leave-one-out cross validation was performed to determine an optimal threshold of high versus low aneuploidy. Candidate thresholds 0.02 to 0.60 in steps of 0.02 were analyzed. A Kaplan–Meier survival model of radiotherapy + ICB versus ICB alone was computed for OS of the patients with tumors harboring an aneuploidy score greater than or equal to each candidate threshold. This process was repeated *n* times, where *n* is the cohort size, leaving out one unique patient in each iteration. For each candidate threshold, the mean log-rank *P* value across the *n* subsets and 95% confidence intervals were computed as well as the mean difference in 12-month survival between the combination radiotherapy + ICB group and the ICB group. The optimal threshold was determined by selecting the candidate threshold at which the difference between the mean *P* value and 12-month survival difference was greatest.

Multiplexed immunofluorescence

Multiplexed immunofluorescence was successfully performed on pretreatment and on-treatment samples from 12 of 15 patients included in the RNA-seq analyses (Supplementary Table 2). A panel of six markers was developed to characterize the tumor microenvironment (Supplementary Table 20). The staining was performed using the Opal Polaris 7-color manual IHC detection kit (Akoya Bio, NEL861001KT) following the manufacturer's instructions. The FFPE tissue slides were deparaffinized by baking at 65°C for 1 h, followed by incubating in xylenes three times for 10 min each. The tissue slides were then rehydrated in a series of ethanol gradients, fixed in 10% NBF solution and rinsed in distilled water. Next, antigen retrieval was performed using a pressure cooker at the low-pressure setting for 15 min in high-pH antigen retrieval buffer. The tissue slides were then blocked in tissue blocking buffer and incubated with the first primary antibody for 1 h at room temperature (20°C) in a humidifying chamber. After a thorough wash with TBST buffer, the slides were incubated with secondary antibody-HRP buffer for 10 min at room temperature. The slides were washed again in TBST buffer and incubated with Opal fluorophores for 10 min at room temperature followed by TBST washes. The pressure cooker treatment was performed again to wash away the excess Opal fluorophore and retrieve antigens for the next primary antibody staining. We repeated the above steps sequentially for each additional antibody. After the final antibody staining, the tissue slides were counterstained with DAPI for 5 min at room temperature, rinsed in water and mounted with ProLong

Diamond Antifade Mountant (Life Technologies, [P36961](#)). After air drying, the multiplexed slides were scanned using an Akoya Vectra Polaris multispectral scanner. The regions of interest were selected using Phenochart software v1.1.0 (Akoya Bio) and image analysis, and tissue and cell segmentation as well as cell phenotype classification was performed using InForm (Akoya Bio) as described in the user manual. CD8⁺ T cells were defined as cells expressing both CD3 and CD8. Tumor cells were defined as cells expressing TTF1 and CK5. Cell densities were calculated by taking the total number of cells in each category across all selected regions of interest and dividing by the sum of the areas of each region.

Statistics and reproducibility

For enrichment analyses, we included only genomic alterations that were present in at least three samples in the cohort or subgroup being analyzed (for example, within a treatment arm) to reduce the likelihood of type I error. When comparing continuous dependent variables, we used a Mann–Whitney test or Kruskal–Wallis test, as appropriate. Splitting variables at the median was performed using the `ntile` function in R. When samples from the same patient were compared, we used a paired test design. For comparisons of two categorical variables, we used Fisher’s exact test. For comparisons of two continuous variables, a Spearman correlation was used. Unless otherwise specified, all tests were performed in R v4.1.1 and were two-tailed. $P < 0.05$ and false discovery rate (FDR) < 0.1 were considered significant. Multiple comparison correction was performed using the Benjamini–Hochberg⁷³ method as implemented in the *qvalue* R package. No statistical method was used to predetermine sample sizes, but our sample sizes are similar to those reported in similar studies^{26,49}. Exclusion criteria are described above. Participants were randomly assigned to sequential or concurrent SBRT + ipi/nivo as described above. The investigators were not blinded during data collection, analysis, experiments or outcome assessment.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

DNA and RNA sequencing data that support the findings of this study have been deposited in the European Genome-Phenome Archive (study ID: [EGAS00001006212](#)). The human lung adenocarcinoma genomic and clinical data were derived from the TCGA Research Network (<http://cancergenome.nih.gov>). Previously published NGS data that were reanalyzed here are available in the supplementary material of²⁴ (UC dataset) and at the following cBioPortal study: https://www.cbioportal.org/study/summary?id=tmb_mskcc_2018 (MSKCC dataset). Source data have been provided in Supplementary Tables 1–16. All other data supporting the findings of this study are available from the corresponding author on reasonable request. Source data are provided with this paper.

Code availability

Code for all relevant genomic analyses is available at the GitHub repository for this manuscript (<https://github.com/lfscurr/COSINR-Analysis>).

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Author contributions

L.F.S. and S.P.P. conceived the study. L.F.S., W.K., M.C., Y.Z., R.H., S.I.G., W.T.T., C.M.L., K.B.P., P.C., S.M., A.N.H., B.C., E.E.V., C.M.B., J.D.P., S.J.C. and S.P.P. acquired the data. L.F.S., C.A.M., W.K., M.C., M.D., T.F.G. and S.P.P. developed the methods. L.F.S., C.A.M., W.K., M.C., Y.Z., A.N.H., M.D., T.F.G. and S.P.P. performed the investigation. L.F.S. and S.P.P. prepared the visualization. L.F.S., C.M.B., J.D.P., R.R.W., S.J.C. and S.P.P. acquired funding. R.R.W., S.J.C. and S.P.P. supervised the work. L.F.S. and S.P.P. initially wrote the manuscript. All authors contributed to the final editing of the manuscript.

Competing interests

S.M. is an advisor for Olympus America, Medtronic, Johnson & Johnson, ERBE, Boston Scientific, Cook and Pinnacle Biologics. E.E.V. has served as an advisor for AbbVie, AstraZeneca, BeiGene, BioNTech, Eli Lilly, ED Serono, Genentech/Roche, GlaxoSmithKline, Merck and Novartis. C.M.B. reports serving in a consulting or advisory role for AbbVie, AstraZeneca, Genentech, Pfizer, Seattle Genetics and Takeda. J.D.P. serves as an advisor for AstraZeneca, Takeda and Genentech, and receives research funding from Bristol Myers Squibb (institution). M.D. reports personal fees from Roche Sequencing Solutions; grants and personal fees from Astra-Zeneca, Illumina and Genentech; personal fees from Novartis, Gritstone Oncology, BioNTech and Boehringer Ingelheim; grants from Varian Medical Systems; other support from CiberMed Foresight Diagnostics; a patent for ctDNA detection issued and licensed to Roche; and patents for ctDNA detection pending and licensed to Foresight Diagnostics. R.R.W. reports having stock and other ownership interests in Boost Therapeutics, ImmVira, Reflexion Pharmaceuticals, Coordination Pharmaceuticals, Magi Therapeutics

and Oncosenesence; serving in a consulting or advisory role for Aettis, AstraZeneca, Coordination Pharmaceuticals, Genus, Merck Serono, NanoProteagen, NKMax America, Shuttle Pharmaceuticals and Highlight Therapeutics, S.L. holds research grants with Varian and Regeneron, and receives compensation (including cost of travel and accommodations and other expenses) from AstraZeneca, Boehringer Ingelheim and Merck Serono. S.J.C. reports participating in the advisory boards for Genentech and AstraZeneca; receiving research support from Bristol Myers Squibb, Merck, EMD Serono and AstraZeneca; and having his spouse who works for Astellas. S.P.P. has patents outside of the submitted work. The remaining authors declare no competing interests.

Additional information

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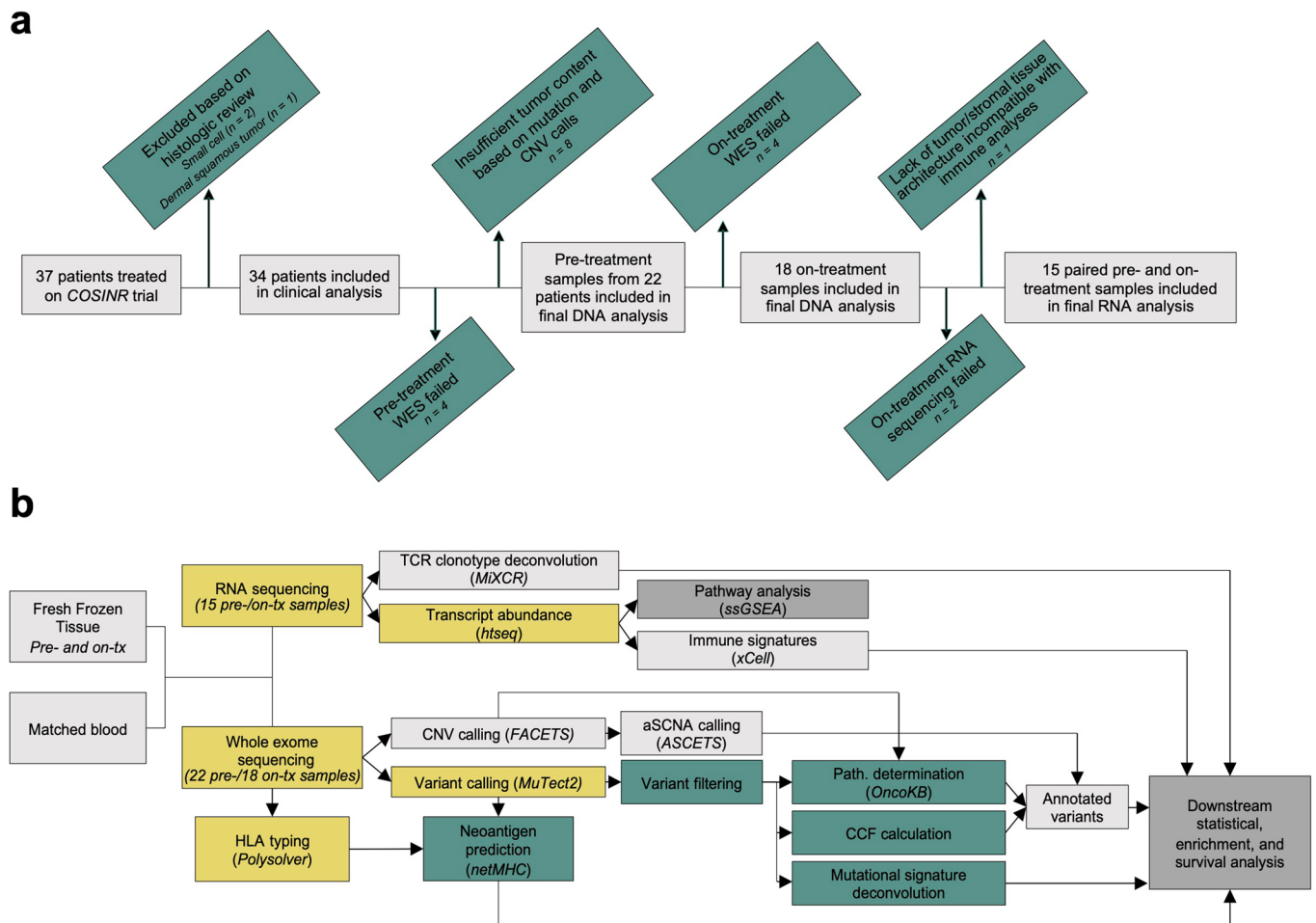
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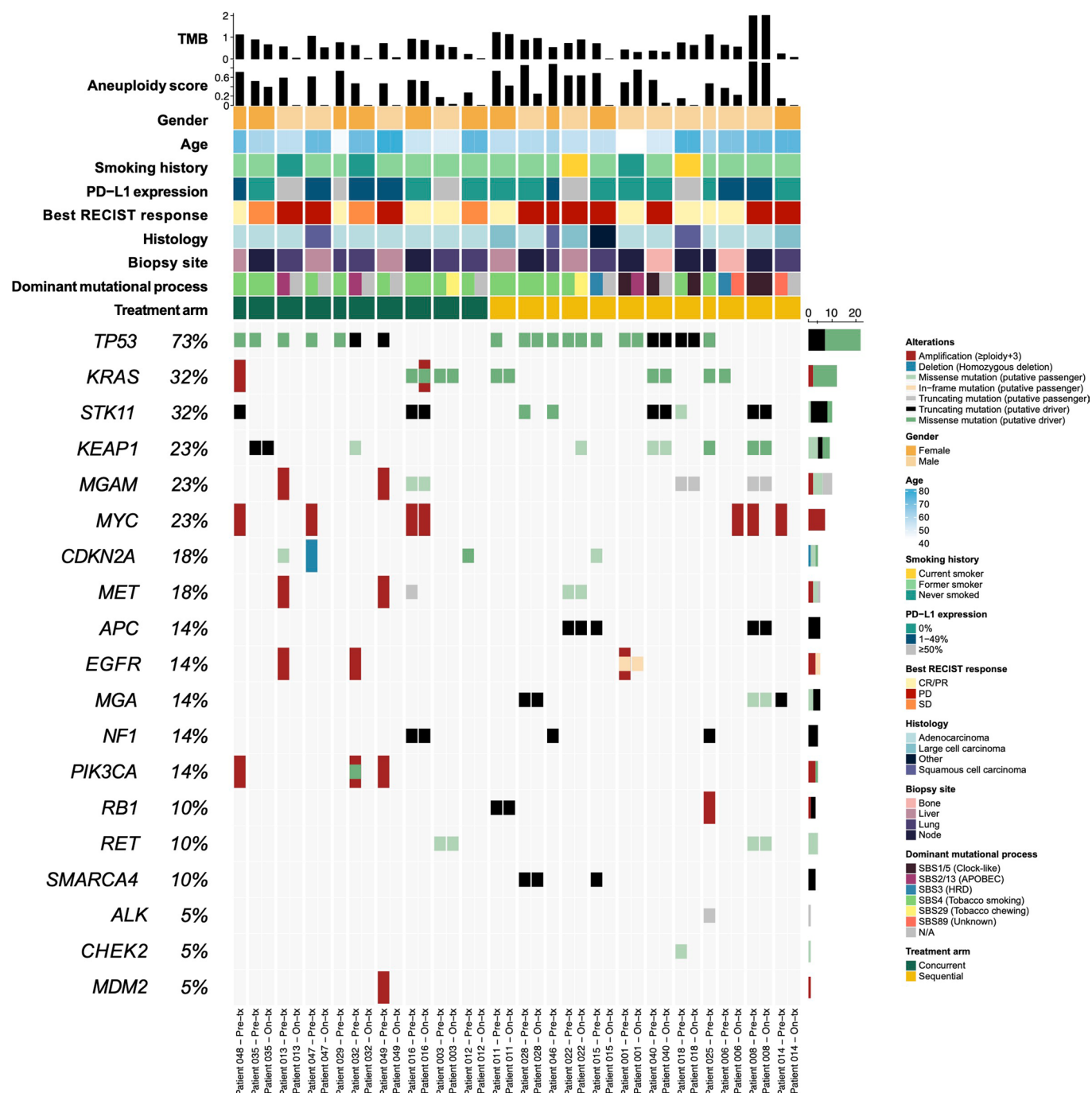
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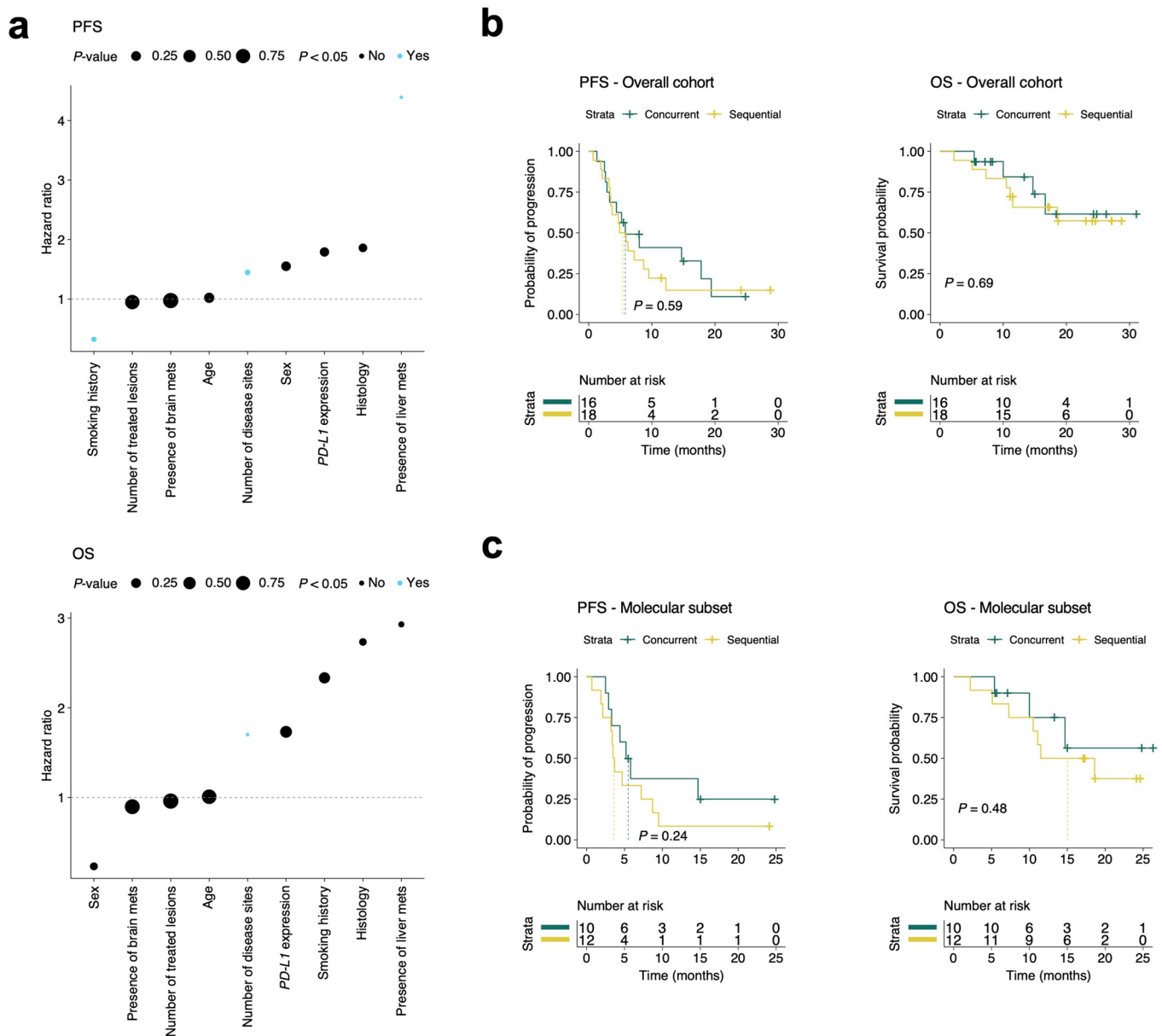


Extended Data Fig. 1 | Consort diagram of patient selection and data analytical framework. a) Patient selection for clinical and genomic analyses. Patients were excluded based on manual pathologic review and inspection of genomic results. **(b)** Schematic of genomic and transcriptomic analysis workflow.



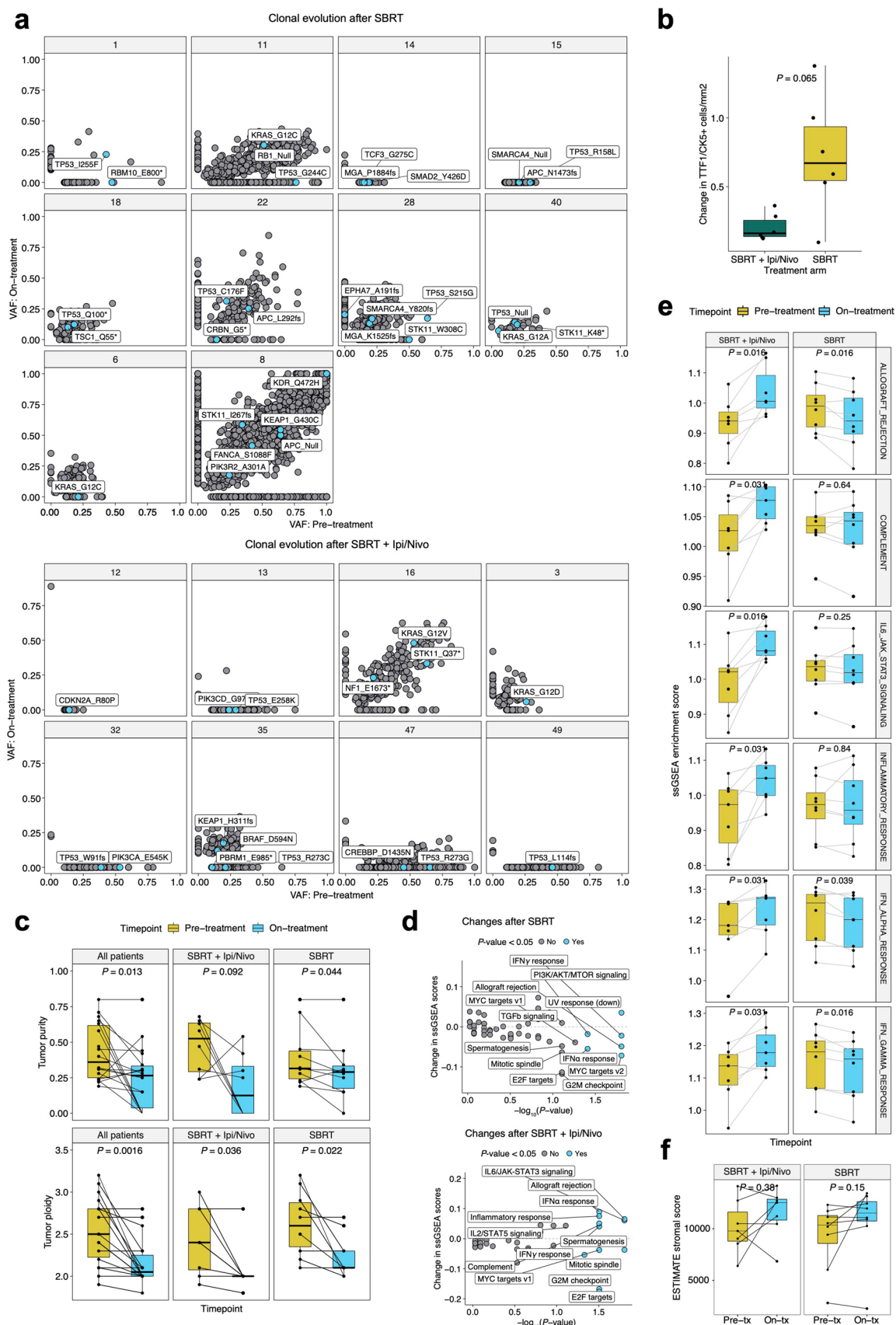
Extended Data Fig. 2 | OncoPrint of COSINR patient cohort. OncoPrint plot of tumors which successfully underwent whole exome sequencing ($n = 40$ samples). Paired samples are adjacent with individual patients separated by larger white breaks. Clinical and pathological data are displayed above variants. Dominant

mutational processes reflect the mutational signature etiology with the largest contribution to the sample's overall mutational profile. Percentages reflect the prevalence of gene alterations in the pre-treatment samples. Bar graphs on the right of the plot reflect the total number of gene alterations across all samples.



Extended Data Fig. 3 | Baseline clinical characteristics of COSINR cohort. (a) Association of clinicopathologic variables with progression-free (top) and overall survival (bottom). **(b)** Progression-free and overall survival of entire cohort by treatment arm. Dotted vertical lines represent median survival; two-sided Log-

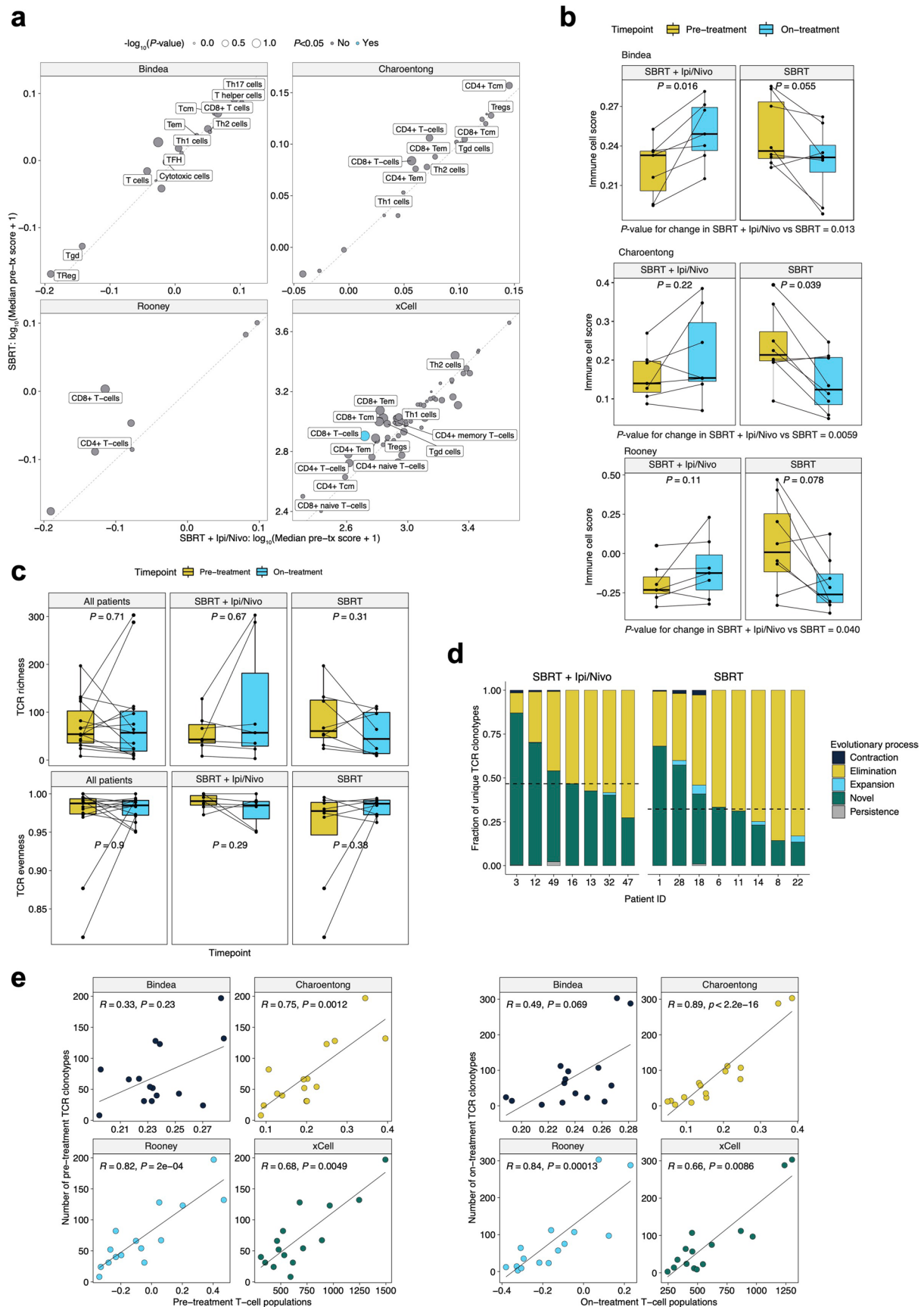
rank test. **(c)** Progression-free and overall survival by treatment arm in the subset of patients used for molecular analysis; two-sided Log-rank test; dotted lines represent median survival ($n = 22$ patients).



Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | Changes in genomic and transcriptomic features on therapy. (a) Clonal evolution of somatic mutations on treatment. Each box corresponds to an individual patient. Axes indicate the variant allele fractions (VAFs) of each mutation (x-axis: pre-treatment; y-axis: on-treatment); oncogenic mutations highlighted in blue; ($n = 18$ patients). **(b)** Differences in on-treatment density of (TTF1+/CK5+) tumor cells as determined by mIF between treatment arms; (SBRT $n = 6$ patients, SBRT+Ipi/Nivo $n = 6$ patients); two-sided Wilcoxon test. The top and bottom edges represent the 1st and 3rd quartiles, respectively; the center line represents the median; whiskers extend to the farthest data points which do not represent outliers (within 1.5x the interquartile range); outliers are plotted as points above and below the box-and-whisker plot. **(c)** On treatment changes in purity and ploidy; two-sided paired Wilcoxon test.

Patient #12 was excluded from the SBRT+ipi/nivo group because the tumor purity could not accurately be determined for the on-treatment sample (SBRT+Ipi/Nivo $n = 7$ patients, SBRT $n = 10$ patients). Boxplot elements are defined in the legend of panel **b**. **(d)** Changes in ssGSEA Hallmark pathway scores on-treatment (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients); two-sided paired Wilcoxon $P < 0.05$ are highlighted in blue. **(e)** Plots illustrating patient-level changes in ssGSEA Hallmark pathways determined to be significantly differentially changed between treatment arms (see Fig. 2d); (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients); two-sided paired Wilcoxon test. Box plot elements are defined in the legend of panel **b**. **(f)** Changes in ESTIMATE stromal score on treatment in each treatment arm; (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients); two-sided paired Wilcoxon test. Box plot elements are defined in the legend of panel **b**.

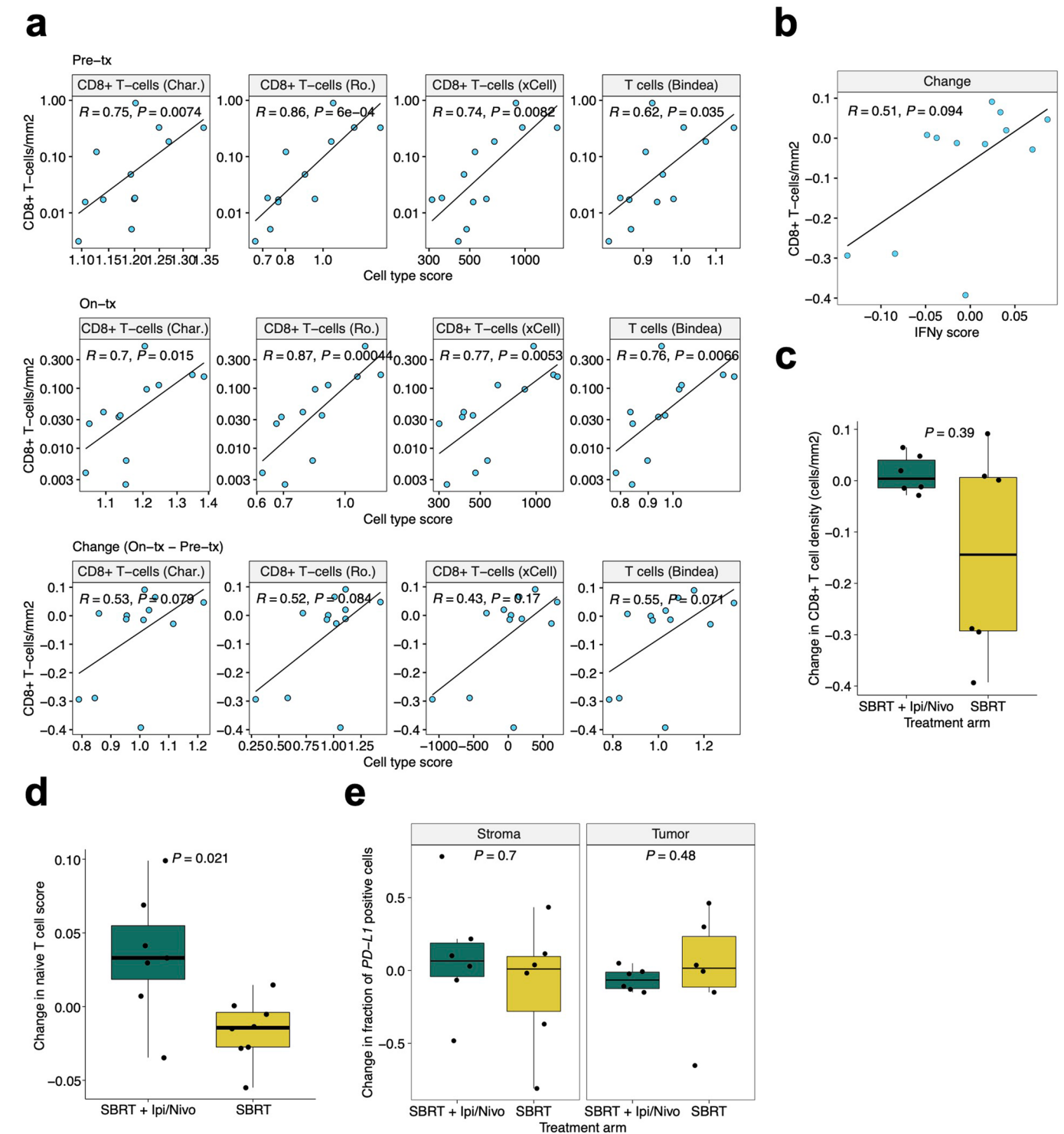


Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | Changes in T cell landscape during treatment.

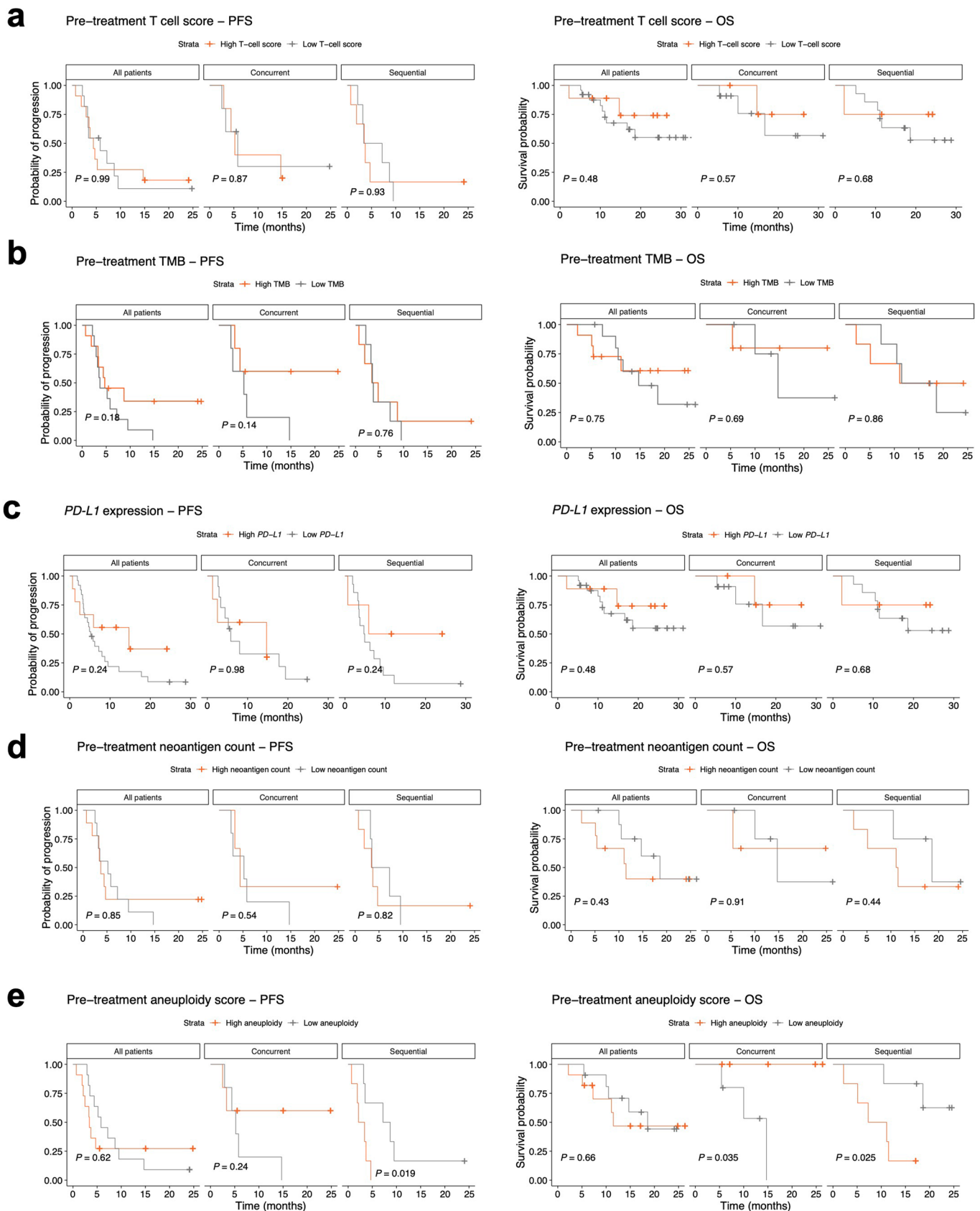
(a) Balance in baseline immune cell signatures across COSINR treatment arms using the four *xCell* signature matrices. Dashed line indicates SBRT = SBRT+Ipi/Nivo. All T-cell associated signatures and any signature with two-sided Wilcoxon $P < 0.05$ are labeled. Blue points are $P < 0.05$ (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients); paired two-sided Wilcoxon signed-rank test. **(b)** Changes in CD8+ T cell populations using the 4 *xCell* signatures; (SBRT $n = 8$ patients, SBRT + Ipi/Nivo $n = 7$ patients). Box plot elements are defined in the legend of Extended

Data Figure 4b. **(c)** Changes in TCR richness and evenness (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients); two-sided paired Wilcoxon test. Box plot elements are defined in the legend of Extended Data Figure 4b. **(d)** Evolution of TCR clonotypes at a per-patient level; horizontal dotted lines represent the median number of novel TCRs per treatment group (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients). **(e)** Two-sided Spearman correlation between pre-treatment (left) and on-treatment (right) CD8+ T cell populations and the number (richness) of TCRs ($n = 15$ patients).



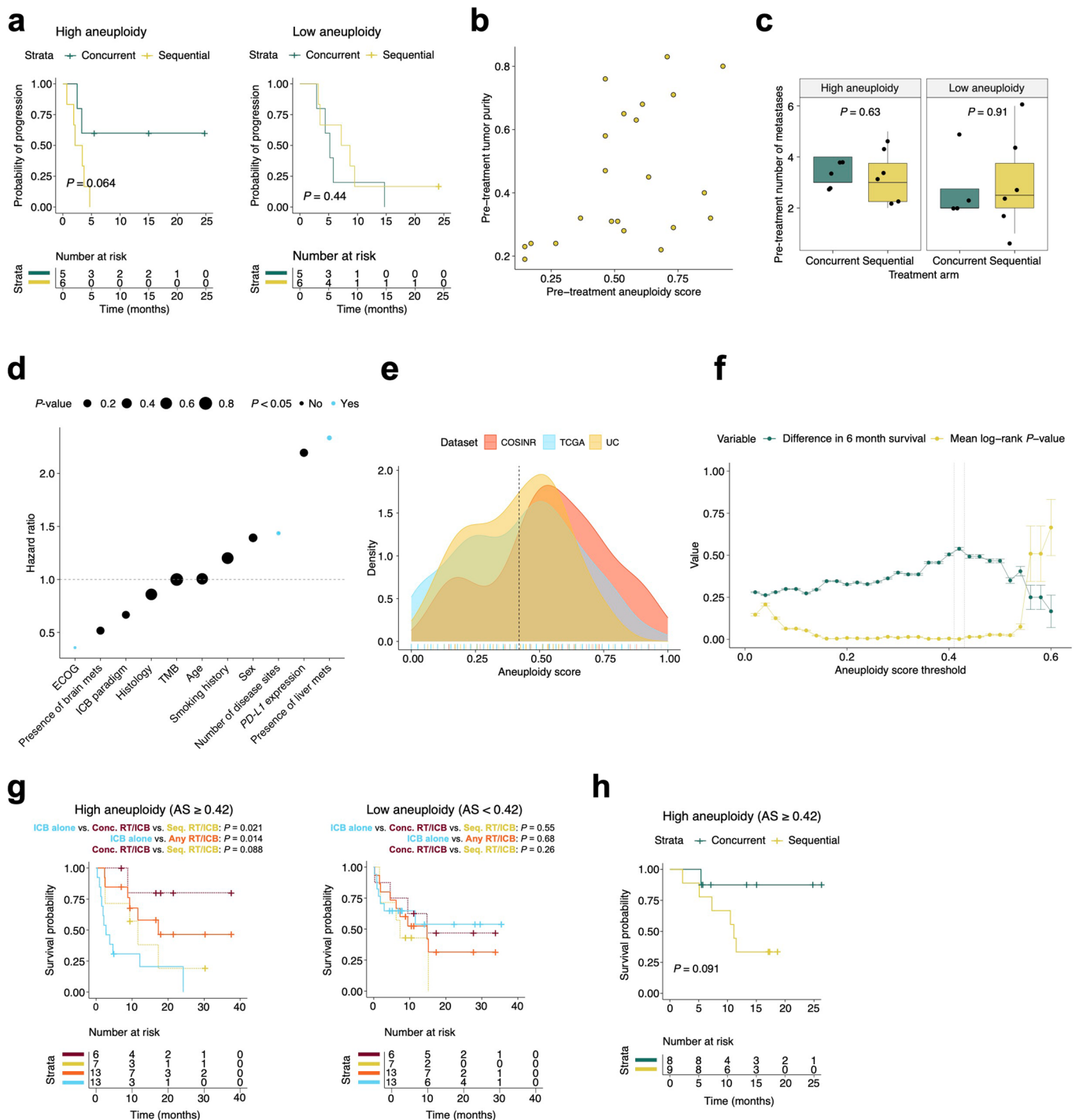
Extended Data Fig. 6 | Integrative T cell characterization using RNA-seq and immunofluorescence. (a) Correlation between xCell immune cell type scores and CD8+ T cell density as determined by mIF; two-sided Spearman correlation (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients). (b) Correlation between change in effector T cell IFN γ signature and change in CD8+ T cell density; two-sided Spearman correlation (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients). (c) Changes in CD8+ T cell density as determined by mIF by treatment arm; (SBRT $n = 6$ patients, SBRT + Ipi/Nivo $n = 6$ patients); two-sided Wilcoxon test. Box plot

elements are defined in the legend of Extended Data Figure 4b. (d) Change in naive T cell gene expression signature by treatment arm; (SBRT $n = 8$ patients, SBRT+Ipi/Nivo $n = 7$ patients); two-sided Wilcoxon test. Box plot elements are defined in Extended Data Figure 4b. (e) Change in the fraction of PD-L1-positive tumor and stromal cells by mIF across treatment arms; (SBRT $n = 6$ patients, SBRT+Ipi/Nivo $n = 6$ patients); two-sided Wilcoxon test. Box plot elements are defined in the legend of Extended Data Figure 4b.



Extended Data Fig. 7 | Association between immunotherapy biomarkers and survival. Association of pre-treatment (a) effector T cell IFN γ signature ($n = 15$ patients), (b) TMB ($n = 22$ patients), (c) PD-L1 expression ($n = 34$ patients), (d)

neoantigen count ($n = 18$ patients), and (e) aneuploidy score ($n = 22$ patients) with progression-free (left) and overall survival (right). Variables were split at the median; two-sided Log-rank test.



Extended Data Fig. 8 | Anaploidy biomarker development in mNSCLC.

(a) PFS for COSINR patients with high aneuploidy score (AS, $\geq$ median) (left) and low AS ($<$ median, right) tumors; two-sided Log-rank test. **(b)** Scatter plot of AS and tumor purity (COSINR, $n = 22$ patients). **(c)** Comparison of number of pre-treatment organ sites by COSINR treatment arm and aneuploidy group ($n = 22$ patients); two-sided Wilcoxon test. Box plot elements are defined in the legend of Extended Data Figure 4b. **(d)** Association of clinical and pathological factors with overall survival in UC cohort ($n = 58$ patients). Variables tested were age, sex (M vs. F), presence of brain or liver metastases, smoking status (ever vs. never), PD-L1 expression ($\geq 50\%$ vs. $< 50\%$), histology (adenocarcinoma vs. other), number of disease sites, TMB, ECOG (0-1 vs. 2-3), and ICB paradigm (monotherapy vs. combination therapy). Variables significantly associated with OS are highlighted

in blue; two-sided Wald test. **(e)** Distribution of AS in COSINR ($n = 22$ patients), UC ($n = 58$ patients), and TCGA ($n = 500$ patients) cohorts; dotted line represents high AS threshold (0.42). **(f)** Selection of optimal high AS threshold based on leave-one-out cross validation analysis; bars: 95% confidence interval; points: mean. Grey lines outline optimal AS threshold (0.42) ($n = 58$ patients). **(g)** Differences in OS in high AS (≥ 0.42) and low AS (< 0.42) groups in UC validation cohort using the derived optimal threshold; two-sided Log-rank test. Dotted maroon and yellow lines represent subdivisions of the RT/ICB treatment group into patients treated with concurrent (maroon) or sequential (yellow) RT + ICB. **(h)** Application of the derived optimal threshold (0.42) to the COSINR cohort (OS); two-sided Log-rank test.

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Software and code

Policy information about [availability of computer code](#)

Data collection

No software or code was used for data collection

Data analysis

All relevant code is available on GitHub: <https://github.com/lfspurr/COSINR-Analysis>
For exome analyses, we utilized Trimmomatic v0.39 [1], BWAmem v0.7.1 [2], samtools v1.1 [3], Picardtools v2.23.8, GATK v4.1.9.0 [4], ANNOVAR v20191024 [5], FOxOG [6], OncoKB v3.1 [7], FACETS v0.6 [8], and ASCETS v1.1.1 [9]. For transcriptome analyses, we used STAR 2.6.1d [10], samtools v1.1 [3], and htseq v0.31.5.

[1] Bolger, A. M., Lohse, M. & Usadel, B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30, 2114–2120 (2014).
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Additional tools used for downstream analysis include:

PolySolver v4.0: Shukla, S. A. et al. Comprehensive analysis of cancer-associated somatic mutations in class I HLA genes. *Nat Biotechnol* 33, 1152–1158 (2015).
 NetMHCpan v4.1: Reynisson, B., Alvarez, B., Paul, S., Peters, B. & Nielsen, M. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. *Nucleic Acids Research* 48, W449–W454 (2020).
 MiXCR v3.0.13: Bolotin, D. A. et al. MiXCR: software for comprehensive adaptive immunity profiling. *Nature Methods* 12, 380–381 (2015).
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 DESeq2 v1.36.0: Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* 15, 550 (2014).
 xCell (May 2020 release): Aran, D., Hu, Z. & Butte, A. J. xCell: digitally portraying the tissue cellular heterogeneity landscape. *Genome Biol* 18, 220–220 (2017).
 cBioPortal: Cerami, E. et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov* 2, 401–404 (2012).
 gsva v1.44.2: <https://www.bioconductor.org/packages/release/bioc/html/GSVA.html>
 biomaRt v2.52.0: <https://www.bioconductor.org/packages/release/bioc/html/biomaRt.html>
 qvalue v2.28.0: <https://www.bioconductor.org/packages/release/bioc/html/qvalue.html>
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- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

DNA and RNA sequencing data that support the findings of this study have been deposited in the European Genome-Phenome Archive (Study ID EGAS00001006212).

The human lung adenocarcinoma genomic and clinical data were derived from the TCGA Research Network: <http://cancergenome.nih.gov/>.

Previously published NGS data that were re-analyzed here are available in the Supplementary Material of ref. 24 (UC dataset) and in the following cBioPortal study: https://www.cbioportal.org/study/summary?id=tmb_mskcc_2018 (MSKCC dataset).

Source data have been provided in Supplementary Data Tables 1-15. All other data supporting the findings of this study are available from the corresponding author on reasonable request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No methods were used to predetermine sample size. We used data from 3 datasets: our COSINR cohort of 37 patients, a previously published cohort of 139 non-small cell lung cancer patients treated with immunotherapy, and an independent cohort of 350 non-small cell lung cancer patients treated with immunotherapy at MSKCC. Sample sizes were based on the public and institutional availability of genomic and clinical data from patients with lung cancer treated with immunotherapy +/- radiation. The sample sizes are comparable to those of similar studies (ref 26 and 49).
Data exclusions	For the COSINR cohort, 22 samples were excluded based on pathologic review and review of sequencing data. Reasons for exclusion included failed sequencing, incorrect tumor types, and insufficient tumor content. The sample exclusion workflow is shown in Extended Data Figure 1a.
Replication	Validation of our findings was performed in two independent cohorts as described above.
Randomization	Patients in the COSINR cohort were randomized prior to treatment to either concurrently or sequentially administered stereotactic body radiotherapy and immunotherapy as part of the phase I COSINR trial. Patients in the University of Chicago validation set were consecutively treated with ICB with or without radiotherapy as per the standard-of-care treatment for patients with metastatic NSCLC. Treatment allocation

was not randomized, but covariates including ECOG performance status, presence of brain metastases, ICB paradigm, histology, TMB, age, smoking history, sex, number of disease sites, PD-L1 expression, and presence of liver metastases were balanced across treatment groups.

Blinding

No blinding was performed because the study examined the timing of RT and ICB; therefore, all patients received the treatments of interest.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

- n/a
- Involved in the study
- ☐ ☒ Antibodies
- ☒ ☐ Eukaryotic cell lines
- ☒ ☐ Palaeontology and archaeology
- ☒ ☐ Animals and other organisms
- ☐ ☒ Human research participants
- ☐ ☒ Clinical data
- ☒ ☐ Dual use research of concern

Methods

- n/a
- Involved in the study
- ☒ ☐ ChIP-seq
- ☒ ☐ Flow cytometry
- ☒ ☐ MRI-based neuroimaging

Antibodies

Antibodies used

- Ki-67: Vendor: BioCare Medical, Cat #: API3156, Clone: MIB-1, Lot #: 112321A, Dilution: Ready to use, Fluorophore: Opal-480

- CD3: Vendor: BioCare Medical, Cat #: CM324, Clone: EP41, Lot #: 032818, Dilution: 1/50, Fluorophore: Opal-520

- CD8: Vendor: Novus Bio, Cat #: NBP232836B, Clone: C8/144B, Lot #: 925-2P170623, Dilution: 1/100, Fluorophore: Opal-570

- PD-1: Vendor: Abcam, Cat #: Ab137132 Clone: EPR4877, Lot #: GR3230470-10, Dilution: 1/300, Fluorophore: Opal-620

- PD-L1: Vendor: Cell Signaling, Cat #: 13684S Clone: E1L3N Lot #: 13, Dilution: 1/2000, Fluorophore: Opal-690

- TTF1+CK5: Vendor: BioCare Medical, Cat #: 425 Clone: 8G7G3/1 EP42 Lot #: 022421A, Dilution: Ready to use, Fluorophore: Opal-425

Validation

Publications containing validation information for each antibody are available on the manufacturer website:

<https://biocare.net/product/ki-67-rabbit-antibody/>

<https://biocare.net/product/cd3-antibody/>

https://www.novusbio.com/products/cd8-antibody-c8-144b_nbp2-32836#datasheet

<https://www.abcam.com/pd1-antibody-epr48772-ab137132.html>

<https://www.cellsignal.com/products/primary-antibodies/pd-l1-e1l3n-xp-rabbit-mab/13684>

<https://biocare.net/product/ttf-1-ck5/>

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics

The COSINR study is a randomized phase I/II trial designed to evaluate the safety and efficacy of combination immune checkpoint blockade (ICB) using ipilimumab + nivolumab (ipi/nivo) plus sequential or concurrent stereotactic body radiotherapy (SBRT) as a first-line treatment for patients with stage IV NSCLC. Age, sex, race, ECOG performance status, smoking history, histology, PD-L1 expression, TMB, number of irradiated disease sites, prior exposure to chemotherapy, prior radiation history, and evidence of brain or liver metastases was not statistically different between the two treatment arms.

Recruitment

Eligibility criteria were previously described (PMID 34500113) and include Eastern Cooperative Oncology Group (ECOG) 0/1 performance status, measurable disease defined by RECIST v1.1, no symptomatic CNS disease, and no autoimmune or immunodeficiency syndromes. Without consideration of PD-L1 expression or TMB, patients were randomized to SBRT to 2-4 metastatic sites with concurrent or sequential (within 7 days) immunotherapy. 37 patients were treated on the phase I component of the COSINR study.

Validation of our findings was performed using a 58 patient subset of 139 patients with NSCLC who underwent next-generation targeted genomic sequencing and were treated with ICB (59% anti-PD-1/anti-PD-L1 and 41% ICB combination) at our institution as previously described. We selected patients with available segmented copy-number data and split patients into three well-defined subsets for further analysis: (1) ICB alone (no radiation; excluded patients who received RT to the brain [radiosurgery or whole brain radiation therapy]; n=30), (2) sequential RT and ICB (interval between treatments of 7-60 days; n=14), and (3) concurrent RT and ICB (interval between treatments of 0 days; n=14). In the sequential and concurrent groups, patients must have received RT to at least one extracranial disease site but were unselected regarding RT dose and target organ site.

No biases were present that could have impacted the results.

Ethics oversight

This study was approved by the University of Chicago Biological Sciences Division IRB Committee (IRB17-0547-CR003). All patients treated on the COSINR trial provided written informed consent. Informed consent was waived for patients in the University of Chicago validation cohort given the retrospective nature of the study.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	NCT03223155
Study protocol	The full COSINR trial protocol is described in: Patel, J. D. et al. Randomized phase I trial to evaluate Concurrent or Sequential Ipilimumab, Nivolumab, and stereotactic body Radiotherapy in patients with stage IV non-small cell lung cancer (COSINR Study). JCO 38, 9616–9616 (2020).
Data collection	A total of 49 patients with stage IV NSCLC were enrolled between September 29, 2017, and October 19, 2019. Overall, 37 patients were eligible for analysis. 18 patients received concurrent treatment and 19 underwent sequential therapy. All clinical data were obtained and stored at the University of Chicago Medical Center.
Outcomes	Clinical evaluation of patients was performed during radiotherapy and with every immunotherapy treatment. Radiographic evaluation was performed using CT scans every 12 weeks, or earlier if clinically warranted. A modified version of RECIST version 1.1 was used to assess overall response, allowing for the inclusion of both irradiated and nonirradiated metastases as target and nontarget metastases. The rest of the RECIST principles were applied. Tumor control was defined as any response other than progressive disease. The Kaplan-Meier method was used to estimate progression-free survival (PFS), defined as the time from SBRT planning to either progression per RECIST criteria or death from any cause. OS was defined as time from SBRT planning to death from any cause.